The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

Anonymous Submission

***Abstract*—Individual differential privacy (iDP) promises each data contributor control over their own privacy: a personal budget (ε_i, δ), chosen by them, enforced for them. The dominant realisation of that promise in deep learning—sampling-based iDP—keeps a single noise multiplier for the whole dataset and turns each budget into a per-record sampling rate, so every contributor's realised membership-inference risk depends on the budget choices of everyone else. If risk is collectively determined, budget choice is a game, and the right question is not whether the mechanism *can* be abused but what *rational* contributors will do with it. We formalise this game as the Privacy Externality Game (PEG), prove that the externality is monotone—tightening any budget raises everyone else's realised risk—and characterise its equilibria. Our headline finding is that the game generically refuses symmetric play: wherever the shared model utility is not yet saturated, every symmetric profile is a local *minimum* of each player's own utility, and stable play is asymmetric—in the canonical instance, 42 of 43 pure equilibria put one player roughly $6.5\times$ looser, with a strictly higher realised risk compensated by the shared gain in model utility. Symmetric play survives only on a knife-edge: a single symmetric pure equilibrium at the budget where utility saturation sets in, which is also the equilibrium with the least aggregate risk. Heterogeneous privacy preferences do not soften the asymmetry; they sort players into protected and exposed roles, with the exposed role going to whoever values privacy least. The asymmetry is a property of the sampling-based realisation, not of iDP itself: the sensitivity-based realisation, replayed through the same game, has a unique and symmetric equilibrium. A recent budget-manipulation attack on iDP emerges in our framework as the coordinated limit of ordinary rational play—and collapses by a factor of 18–53 when the target is allowed to best-respond. Throughout, risk is priced mechanism-specifically: with the strongest adversary the deployed mechanism admits, not the loose mechanism-agnostic (ε, δ) bound; a measured shadow-model instance locates the boundary of the phenomenon, which vanishes where realised attack power sits at the measurement floor. Because no report-based fix exists (we show the mechanism is not strategyproof), mitigation must change the contract itself; a $(\varepsilon_i, \delta, \Delta)$-iDP contract provably caps the externality any coalition can inflict.**

***Index Terms*—Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.**

I. Introduction

The right to privacy is personal, and privacy preferences are heterogeneous: a clinical patient and a casual app user contributing records to the same model place very different values on not being identifiable from it. Classical differential privacy (DP) [1] ignores this heterogeneity—a single budget (ε, δ) covers every record. individual differential privacy (iDP) [2]–[4] restores it: each record receives its own contract (ε_i, δ), chosen to match its owner's preference. The promise is individual control. This paper is about how that promise fares once the contributors holding those budgets are allowed to choose them rationally.

The promise meets reality in the mechanism that delivers it. The dominant deep-learning realisation of iDP, the sampling-based mechanism of [3], keeps a *single* noise multiplier σ for the whole training run and translates each record's budget into a personal Poisson sampling rate p_i, calibrated so that every contract holds at equality under a fixed expected batch size. The shared σ and the fixed batch size couple every sampling rate to every budget: when one contributor tightens their budget, the mechanism must re-solve, and everyone else's sampling rate—and with it their *realised* membership inference (MIA) risk—moves. [5] demonstrated this empirically and weaponised it: an adversary controlling other records' budgets can inflate a target's MIA susceptibility while every formal (ε_i, δ) guarantee continues to hold. Individual privacy risk, under this realisation, is collectively determined.

This is a game. A choice whose consequences are determined by everyone's choices is, by definition, the subject of game theory. Prior work treats the collective determination of iDP risk as an attack surface: something a malicious coalition exploits against a victim [5]. We ask the question that comes before any attack: *what do ordinary, self-interested contributors do when they understand the coupling?* Each player values the shared model and dislikes their own realised MIA risk—priced, throughout, mechanism-specifically: the most any adversary can extract from the mechanism actually deployed, tighter than the mechanism-agnostic (ε, δ) worst case and above any concrete attack (Section II-C); each knows that tightening their budget raises the others' realised vulnerability while degrading the model for all. We call the externality at the heart of this trade-off the *privacy externality* and the resulting normal-form game the Privacy Externality Game (PEG). Its equilibria—not any adversary—are our object of study.

The answer is unsettling. Equal protection is available: the mechanism recomputes every player's realised vulnerability from the whole budget vector at every profile, and profiles at which those vulnerabilities coincide exist throughout the feasible region. But rational play almost never rests at one. Wherever the shared model utility is not yet saturated, each player's utility has a local *minimum* in their own action at any symmetric candidate, so conforming is the one thing rational

players will not do; symmetric play survives at exactly one knife-edge budget, where saturation sets in. Everywhere else the game behaves like the classical Hawk–Dove game [6]: the roles are strategic substitutes, deviations from symmetry amplify rather than damp back, one player ends with a strictly higher realised MIA advantage than the other, and who ends up exposed is decided by equilibrium selection, not by anyone's contract. When preferences are heterogeneous the roles stop being arbitrary—the equilibrium *sorts* players, assigning the exposed role to whoever values privacy least. Risk under sampling-based iDP is not merely interdependent, as prior work observed; at equilibrium it is systematically *unequal*.

Contributions. The main contributions of our work are as follows.

- We formalise sampling-based iDP budget choice as the n-player Privacy Externality Game and prove the externality is monotone: tightening any budget raises every other player's realised MIA advantage, shrinking their slack below the DP ceiling.
- We prove the diagonal is a trough of own utility wherever the shared utility is unsaturated ($\sigma \geq \sqrt{3}$ for the canonical V), so the symmetric game has no interior symmetric pure equilibrium there, and budgets are strategic substitutes, so deviations amplify. Where saturation wins, we locate the single symmetric pure equilibrium—it is also the welfare-best equilibrium—and heterogeneous preferences sort the exposed role of the asymmetric equilibria onto whoever values privacy least.
- We show the asymmetry is mechanism-specific—the sensitivity-based realisation yields a unique, symmetric equilibrium—and recover the budget-manipulation attack of [5] as rational coalition play, which collapses 18–53$\times$ against a target that best-responds.
- We quantify the welfare cost of equilibrium play (the price of externality-vulnerability, PoEV), prove that budget declaration is not strategyproof (no reporting discipline can repair the mechanism from inside), and prove that a $(\varepsilon_i, \delta, \Delta)$-iDP contract caps at Δ the risk movement any coalition can inflict.

Implication. The attack literature on iDP frames excess vulnerability as something an adversary inflicts. Our equilibrium analysis says the adversary is optional: asymmetric vulnerability is what rational play produces on its own, and the coalition attack is merely its coordinated limit. Mitigation therefore cannot proceed by detecting attackers—there may be none—but must change the contract so that the game itself stops admitting harmful equilibria.

II. Background and Related Work

A. Differential privacy

A randomised mechanism $\mathcal{M}$ is (ε, δ)-differentially private ((ε, δ)-DP) [1] if, for all adjacent datasets $D \sim D'$ and all measurable output sets S,

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{1}$$

differentially private stochastic gradient descent (DP-SGD) instantiates Eq. (1) for deep learning: per-example gradients are clipped, their sum is perturbed with Gaussian noise $\mathcal{N}(0, \sigma^2 I)$, and Poisson subsampling—each record enters each batch independently with probability p—amplifies the per-step guarantee before composition over training iterations [7]. A record's protection is therefore set by two dials: the noise multiplier σ, shared by all records, and its own sampling rate p.

B. Individual differential privacy

A single (ε, δ) imposes one privacy level on contributors whose dispositions demonstrably differ: three decades of surveys following Westin consistently segment any surveyed population into privacy fundamentalists, pragmatists, and unconcerned [8], and the split persists among users of emerging health technologies [9], even though the categories predict individual behaviour only loosely [10]. We write $\lambda_i > 0$ for the utility/privacy weight contributor i attributes to their data: how much they value a unit of their own realised leakage relative to a unit of model utility. The survey evidence says λ_i varies widely within any one dataset—a patient with a stigmatised diagnosis and a healthy volunteer in the same clinical study, or a rare-disease clinic and a large research hospital in the same cross-silo federation [11]–[13]. iDP [2] operationalises this heterogeneity, giving record i its own budget: $\mathcal{M}$ is (ε_i, δ)-iDP if for every i, every pair $D \sim_i D'$ differing only in record i, and all S,

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon_i} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{2}$$

Realisations fall into two families [3], [4]. *Accounting* realisations track each record's cumulative expenditure and filter it out once its budget is exhausted; the progressive record removal causes catastrophic forgetting and poor utility [3], [4]. *Assignment* realisations instead fix per-record training parameters up front: *sensitivity-based* iDP assigns each record a personal clipping norm, *sampling-based* iDP a personal Poisson sampling rate under one shared noise multiplier. Sampling-based iDP attains the better empirical utility of the two [3], [5]; we adopt the formulation of [3] and use the sensitivity-based variant as a control (Section V-B).

Definition 1 (Sampling-based iDP-SGD [3])**.** Let the dataset of size M partition into G *privacy groups* $\{G_1, \ldots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ)-DP budget. The mechanism trains by Poisson-subsampled DP-SGD for I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \ldots, p_G$ are chosen such that

$$\text{for every } g, \text{ the } I\text{-fold } (p_g, \sigma)\text{-sub-}$$
$$\text{sampled Gaussian is } (\varepsilon_g, \delta)\text{-DP at equality;} \tag{3}$$

$$\sum_{g=1}^{G} \frac{|G_g|}{M}\, p_g \;=\; \frac{B}{M}, \tag{4}$$

where B is the desired expected batch size. We refer to $(\sigma, p_1, \ldots, p_G)$ as the *mechanism solution* for the budget vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_G)$.

We account for the subsampled-Gaussian composition in Rényi differential privacy (RDP) [7], [14] for tight composition, but state each contract in (ε_i, δ) form, following [3], [5]. The budget (ε, δ) is the formal quantification of a record's information leakage [1]; it is the basis of the contract between a data contributor and the party that processes their data [5], and the form in which guarantees are deployed and communicated in practice [15].

C. Membership inference, privacy profiles, and slack

A contract (ε, δ) is a lossy description of a mechanism's leakage. In Eq. (1) the two parameters compensate for each other—a larger ε lets the same mechanism claim a smaller δ—so a fixed mechanism satisfies Eq. (1) not at one pair but along a whole curve: its *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \geq 0\}$, where $\delta(\varepsilon)$ is the smallest δ at which Eq. (1) holds [16]. The profile describes the leakage completely; the contract is a single point on it, and mechanisms through the same point can have very different profile shapes [5], [17], [18]. The risk a contributor actually faces is determined by that shape, not by the calibration point.

To make risk operational we use MIA: an adversary observing the mechanism's output tests whether a target record was in the training set [19], [20]. DP is itself a hypothesis-testing guarantee, and the profile is equivalent to a *trade-off curve* $T(\alpha)$: the least type-II error at type-I level α that any test—hence any MIA adversary—can achieve [21]. The adversary's best operating point defines the advantage,

$$\mathrm{Adv}(\mathcal{M}) \;=\; \sup_{\alpha \in [0,1]} \big(1 - \alpha - T(\alpha)\big), \tag{5}$$

the largest gap between true-positive and false-positive rate. We compute the trade-off curve of the I-fold (p, σ)-subsampled Gaussian exactly by numerical accounting [22] and read the advantage off Eq. (5).

The advantage hierarchy. The paper moves between three notions of vulnerability, one per level of knowledge about the mechanism.

- $A_{\varepsilon,\delta}$ *(contract level, mechanism-agnostic).* Knowing only the contract, the worst case over every mechanism it admits is $A_{\varepsilon,\delta} := (e^\varepsilon - 1 + 2\delta)/(e^\varepsilon + 1)$ [17], [23]: no (ε, δ)-DP mechanism, whatever its form, can leak more. As a worst case over profiles of every shape, it is generally loose for any particular mechanism.
- A_{PLD} *(mechanism level).* Knowing the deployed mechanism—the actual sampling rates, Gaussian noise, and composition, resolved by the exact privacy-loss distribution (PLD) accountant, whence the name—its trade-off curve gives the advantage Eq. (5) of the *optimal* adversary: a true upper bound on what any attacker, with any decision rule, can extract from that mechanism. This is the vulnerability our game prices; player i's vulnerability $A_{\mathrm{PLD},i}$ is always this quantity.
- A_{LiRA} *(attack level).* The realised success of a concrete attack against one trained model. The likelihood-ratio attack (LiRA) [20] instantiates the membership hypothesis test with shadow models and is currently among the most powerful attacks that are realistic and applicable—the practical means by which an adversary leverages the mechanism's information leakage. Realised per-record vulnerability at a fixed budget is highly heterogeneous [24].

The three levels are ordered:

$$A_{\varepsilon,\delta} \;\geq\; A_{\mathrm{PLD}} \;\gtrsim\; A_{\mathrm{LiRA}}. \tag{6}$$

The first inequality holds by construction—level 2 evaluates one mechanism inside the level-1 worst case. Its gap,

$$\mathrm{slack} \;:=\; A_{\varepsilon,\delta} - A_{\mathrm{PLD}} \;\geq\; 0,$$

is the headroom between what the contract permits and what the mechanism actually leaks; the Δ-divergence of [17] makes this gap decision-theoretic, quantifying as an approximate Blackwell dominance the worst-case excess vulnerability of running one mechanism in place of another at the same contract. The second inequality is not a theorem: an attack that is not information-theoretically optimal should achieve no more than the optimal adversary [25], but the trade-off curve is computed for a model of the implemented mechanism—training dynamics, finite precision, and implementation detail are abstracted away—so we treat $A_{\mathrm{PLD}} \gtrsim A_{\mathrm{LiRA}}$ as an empirical regularity. Figure 1 makes the first two levels concrete: six subsampled-Gaussian mechanisms whose profiles pass through one shared contract point realise trade-off curves and advantages differing by orders of magnitude, all far below the ceiling $A_{\varepsilon,\delta}$ (orange bar in panel (c)). Level-3 quantities carry an explicit LiRA superscript.

Where the externality appears. In sampling-based iDP the budget vector selects the mechanism, and Fig. 1 shows how much that selection matters. The shared σ in (3) is the channel: for any candidate σ, the largest sampling rate that meets group g's budget increases in σ (more noise, more room to sample), and (4) then pins σ as the unique value at which the rates sum to B. When any group tightens its budget the mechanism re-solves, and every other group's (p_g, σ)—its trade-off curve, its advantage, its slack—moves along exactly the family of mechanisms in Fig. 1 while every contract point stays put. Record i's realised risk is thus a function of *everyone's* choices, while its contract mentions only ε_i. Fixing the expected batch size is only one way to aggregate the per-record budgets into a shared σ; the externality is not specific to it but is inherent to any informed choice of a shared noise multiplier [5], so the game we build applies to any such rule. [5] demonstrated the stakes: adversarial budget choices inflate the MIA susceptibility of 62% of targeted records without touching any contract, and a $(\varepsilon_i, \delta, \Delta)$-iDP clause capping the slack was proposed as the remedy. We take the complementary route: rather than restricting the mechanism, we explain the realised excess vulnerability as the equilibrium of self-interested play under the current contract—and return to the Δ-clause as mitigation in Section VII.

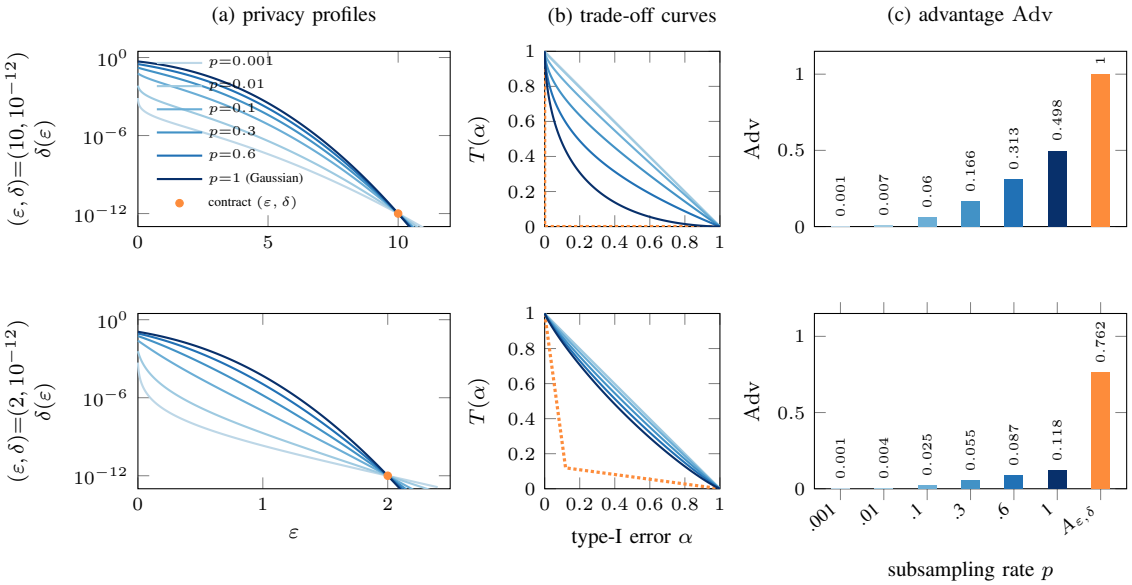


Fig. 1: **The advantage hierarchy at two contract points** (top: $\varepsilon = 10$, $\delta = 10^{-12}$, the paper's canonical setting; bottom: $\varepsilon = 2$, $\delta = 10^{-12}$). Six Poisson-subsampled Gaussian mechanisms (blues), each calibrated by exact PLD accounting so that its privacy profile passes through the row's contract point (orange dot). *(a)* Away from the shared point, the profiles differ by orders of magnitude. *(b)* The exact trade-off curves $T(\alpha)$ [21]; the thin diagonal is no leakage, the dotted orange envelope $f_{\varepsilon,\delta}$ the mechanism-agnostic worst case. *(c)* The realised advantages A_{PLD} sit far below the ceiling $A_{\varepsilon,\delta}$ (orange bar): up to 0.50 versus ≈1 at the top contract, 0.12 versus 0.76 at the bottom. The contract alone neither determines nor tightly bounds the realised risk.

D. Strategic privacy: games, externalities, incentives

A normal-form game $\langle N, \Sigma, U\rangle$ consists of players $N = \{1,\dots,n\}$, action sets $\Sigma = \{S_1,\dots,S_n\}$, and payoffs $U = \{u_1,\dots,u_n\}$. A Nash equilibrium (NE) is a profile at which no player gains from unilateral deviation [26]; a best response (BR) maximises a player's utility against fixed opponent play; the *price of anarchy* compares worst-equilibrium welfare to the social optimum [27]; in a Stackelberg game a leader commits before followers best-respond. Two classical templates organise our results. *Hawk–Dove* [6] is the canonical symmetric game whose pure equilibria are all asymmetric: someone must yield, and the equilibrium—not anyone's preference—decides who. Shared-resource games [28] show how a common constraint (here, the shared σ under a fixed batch size) turns each player's private choice into an externality [29] on all others.

Privacy games. [30] introduced the closest predecessor to our game: a two-player collaborative-learning game in which each player chooses a privacy parameter and the *Price of Privacy* measures the accuracy lost at equilibrium. In their mechanism, however, a player's privacy choice affects only their own *privacy* and both players' *accuracy*: the privacy cost is separable, which makes their game a potential game [31] and its analysis a one-dimensional minimisation. In sampling-based iDP every player's privacy choice moves every other player's *privacy*—a qualitatively stronger coupling that destroys the potential structure (Proposition 12). Earlier game-theoretic privacy work studies different couplings: [32] model regression with private features as a non-cooperative game, [33] study non-monetary incentives in data analytics, and [34] model the learner-vs-perturber conflict as a Stackelberg game. None exhibit the privacy-to-privacy externality of a shared sampling mechanism. On the mechanism-design side, selling privacy as a good with heterogeneous valuations traces to [35], with truthfulness impossibilities for unbounded privacy valuations [36], [37]; we show in Section G why these positive results do not rescue budget declaration in the PEG.

Where the game lives. The PEG lives in systems that individualise privacy budgets and solve for a shared noise multiplier from the resulting budget vector. The budgets can be record-specific—a training instance running sampling-based iDP over its own dataset [3]—or silo-specific, as in cross-silo federated learning with personalised budgets [11]–[13], [38], where the silos—hospitals in a consortium, banks in a fraud-detection coalition—are exactly the separately identifiable entities holding heterogeneous preferences. A separate literature designs incentives on top of privacy-heterogeneous federated learning [39]–[42]; our equilibrium characterisation is the baseline any such scheme must improve upon—it is what happens when no incentive is offered.

III. The Privacy Externality Game

We formalise the strategic situation faced by data contributors under sampling-based iDP as a normal-form game on a continuous action space. The formalisation deliberately follows the two-player collaborative-learning template of [30], with one

central modification: the privacy cost is no longer separable across players, because the mechanism couples it.

A. Players, actions, and utilities

Definition 2 (Privacy Externality Game)**.** Fix a common $\delta > 0$, an expected batch size B, an iteration count I, and a partition of the dataset into n groups of sizes $|G_1|, \ldots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where:

- $N = \{1, \ldots, n\}$ is the set of players. We adopt the federated-silo granularity of [5]: player i is the entity controlling group i. Each player picks a single scalar budget for their group; per-record budget distributions within a group are an extension we discuss in Section VIII.
- Each player's action set is $\Sigma_i = [\varepsilon^{\min}, \varepsilon^{\max}] \subset \mathbb{R}_{>0}$.
- Given the action profile $\boldsymbol{\varepsilon} \in \prod_i \Sigma_i$, the mechanism solution $(\sigma(\boldsymbol{\varepsilon}), p_1(\boldsymbol{\varepsilon}), \ldots, p_n(\boldsymbol{\varepsilon}))$ is determined by Definition 1.
- Each player's utility is

$$u_i(\boldsymbol{\varepsilon}) \;=\; V(\sigma(\boldsymbol{\varepsilon})) \;-\; \lambda_i \cdot A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}), \tag{7}$$

  where $\lambda_i > 0$ is the player's utility/privacy weight of Section II-B, pricing a unit of realised MIA advantage on their own data against a unit of model utility.

Definition 3 (Vulnerability function)**.** The *vulnerability function* of player i is $A_{\mathrm{PLD},i} \colon \prod_i \Sigma_i \to [0,1]$, $\boldsymbol{\varepsilon} \mapsto \mathrm{Adv}\big(\mathcal{M}_{p_i(\boldsymbol{\varepsilon}),\, \sigma(\boldsymbol{\varepsilon}),\, I}\big)$, the tight analytical MIA advantage (5) of the I-fold (p_i, σ)-subsampled Gaussian.

In the hierarchy (6), $A_{\mathrm{PLD},i}$ is a level-2 object: the mechanism-specific supremum over all adversaries against the mechanism actually solved for at the profile—a true bound on what any attacker can extract there, unlike the mechanism-agnostic level-1 ceiling written in player i's contract, and unlike the outcome of any particular attack. It is the advantage measure of [5] transferred unchanged to a strategic setting. Its dependence on $\boldsymbol{\varepsilon}$ runs entirely through (p_i, σ), both of which the mechanism's constraints (3) and (4) determine from the whole budget vector; this coupling is the externality.

The model-utility term. For V we use

$$V(\sigma) \;=\; \frac{1}{1+\sigma^2}, \tag{8}$$

monotone-decreasing in the noise multiplier and bounded in $[0,1]$. The shape is motivated by the excess-empirical-risk bound of [43] for convex DP-ERM, which under the moments accountant of [7] translates into a σ^2 dependence of the excess risk (Section C); in the potentially non-convex deep-learning regime we use it as a directional proxy, consistent with the monotone accuracy degradation reported empirically [3], [7], [44]. None of our results hinge on this choice: Proposition 20 shows that every load-bearing statement holds for any V that is continuous, strictly decreasing, and bounded on the feasible noise range, and Section C-A verifies numerically that the equilibrium structure is unchanged under six qualitatively different shapes.

Assumption 4 (Participation)**.** A profile $\boldsymbol{\varepsilon}$ is *feasible* iff the mechanism solution satisfies $p_i(\boldsymbol{\varepsilon}) > 0$ for every player. Profiles with $p_i = 0$ correspond to a player whose data never enters training; their contract holds trivially and we exclude such profiles, restricting each action set to the participating region.

B. Contracted versus excess cost

The slack of Section II-C, instantiated per player, separates the vulnerability into a contracted part and a played part:

$$A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) \;=\; A_{\varepsilon_i,\delta} - \mathrm{slack}_i(\boldsymbol{\varepsilon}), \qquad \mathrm{slack}_i(\boldsymbol{\varepsilon}) \;\geq\; 0. \tag{9}$$

The two terms have different owners. The level-1 ceiling $A_{\varepsilon_i,\delta}$ depends on player i's budget alone—it is written in their contract—while the slack is a function of the entire profile. The externality therefore lives exclusively between levels 1 and 2 of (6): when others tighten their budgets, slack_i shrinks and $A_{\mathrm{PLD},i}$ climbs toward the ceiling (Proposition 18); the Δ-clause of Section VII caps exactly this movable term.

C. Coalition variant

Definition 5 (k-coalition Stackelberg PEG)**.** Fix a target $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k-coalition Stackelberg PEG is the two-stage game in which (i) the coalition jointly commits to an action vector $\boldsymbol{\varepsilon}_C$ maximising the augmented utility

$$\sum_{i \in C} u_i(\boldsymbol{\varepsilon}) \;+\; \kappa\, A_{\mathrm{PLD},t}(\boldsymbol{\varepsilon}), \tag{10}$$

with *malice weight* $\kappa \geq 0$ on the target's vulnerability [30], and (ii) the remaining players best-respond. $\kappa = 0$ recovers the benign PEG; $\kappa \to \infty$ recovers the budget-manipulation threat model of [5].

D. Equilibrium concepts and information

We use pure Nash equilibrium on the feasible region as the primary solution concept. Existence on the continuous action space is not automatic—u_i is not concave in ε_i, so the classical quasi-concavity route [45], [46] does not apply; Section IV-D gives an order-theoretic existence result in the practically relevant regime. Numerically we enumerate pure NEs by brute force on a dense log-spaced grid $\mathcal{A}_K \subset [\varepsilon^{\min}, \varepsilon^{\max}]$ (a finite game, so a mixed NE exists in any case [26]) and use fictitious play [31] where enumeration is out of reach.

Remark 6 (Awareness is information, not utility)*.* Every player has the same utility (7); what separates them is the level of the hierarchy (6) at which they can price its cost term. A *nominal-information* player sees only their own contract and therefore prices at level 1, taking the ceiling $A_{\varepsilon_i,\delta}$ as their cost. That ceiling depends on their own budget alone, so their best response ignores $\boldsymbol{\varepsilon}_{-i}$: they play the separable Price-of-Privacy game of [30] unchanged. An *aware* player can compute $A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon})$ from the public mechanism parameters, prices at level 2, and plays the PEG. The distinction matters for the attack analysis of Section VI: the known attacks succeed against nominal-information targets, and awareness alone turns out to be a substantial defence.

IV. Equilibrium Analysis: Equal Protection Is Unstable

As orientation before the details, the section's argument in brief: the externality has a definite sign—tightening one player's budget raises every other player's PLD-advantage at their unchanged contracts; symmetric play is unstable wherever the shared utility is unsaturated—one knife-edge symmetric equilibrium survives where it is not—so equilibrium protection is generically unequal; preference heterogeneity decides in whose favour; equilibria exist in the practically relevant regime, even though the standard potential-based existence arguments do not apply; and coalition play recovers the known attack. The subsections develop these claims in this order; full proofs are deferred to Section B, and the short ones stay inline. Each claim is checked against numerical evidence in place, so that the analysis and its verification read as one argument.

Numerical pipeline. All numerical evidence below is analytical—no model is trained—and comes from one pipeline. We solve the mechanism of Definition 1 by bisecting on σ until (4) is met, with each per-group rate chosen to make (3) bind under the Rényi accountant [14]; profiles where any p_i falls below a participation threshold are flagged infeasible (Assumption 4, Section A). Each realised $A_{\mathrm{PLD},i}$ is then computed exactly by PLD accounting [22]—the level-2 advantage of (6), not the level-1 ceiling. The canonical action grid is log-spaced with $K = 200$ points on $[0.5, 64]$ at $\delta = 10^{-12}$; unless noted otherwise, groups are symmetric with $|G_g| = 5000$, $B = 128$, 5 training epochs, $\lambda_i = 1$, and $V(\sigma) = 1/(1+\sigma^2)$, mirroring the empirical iDP scenarios of [3], [5].

A. The externality is monotone

Player i's risk depends on exactly two quantities, (p_i, σ), and the mechanism sets both from the whole budget vector. The first lemma fixes the *direction* in which the other players' choices move that pair.

Lemma 7 (Privacy externality on the feasible region)**.** *Consider the sampling-based iDP mechanism of Definition 1 on a feasible profile $\boldsymbol{\varepsilon}$ at which constraints* (3) *bind for every group and* (4) *binds at B. Then for every pair $i \neq j$,*

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial p_i}{\partial \varepsilon_j} \;<\; 0. \tag{11}$$

When player j tightens their budget, the mechanism adds noise and samples everyone else more.

Proof idea. The externality travels through what all players share: the batch and the noise. Amplification makes the sampling rate at which group g exhausts its budget, $p_g(\sigma, \varepsilon_g)$, increase in that budget and in the shared σ [16], so the expected batch is strictly increasing in σ and (4) pins a unique σ. When player j tightens their budget, they relinquish part of the fixed batch; the solver can restore B only by raising σ, and the extra noise raises every other group's rate. Player i's mechanism has moved—in both p_i and σ—though ε_i never did. Full proof in Section B-A. □

The lemma does not yet determine the sign of the externality: when player j tightens their budget, player i is sampled more but under more noise, and the two effects oppose each other. The next two results settle the sign, and both rest on the same observation. Player i's contract holds ε_i fixed and their budget constraint (3) constraint binds at equality; this one equation ties p_i to σ, confining i's operating point to the single curve in the (p_i, σ)-plane on which the mechanism's leakage equals ε_i. To leading order this curve is the line $p_i \propto \sigma$: at Rényi order α, with $\rho_g(\varepsilon_g, \delta)$ the group's RDP budget [14], (3) reads $p_g = \sigma\sqrt{\rho_g(\varepsilon_g,\delta)}/\sqrt{2I\alpha}$, and summing these rates across groups under (4) gives the aggregate identity

$$\sigma \cdot \sum_g |G_g| \sqrt{\rho_g(\varepsilon_g, \delta)} \;=\; B\sqrt{2I\alpha}. \tag{12}$$

Tightening ε_j raises σ (Lemma 7) and moves player i along this line toward larger σ. The sign of the externality thus reduces to a single question: does $A_{\mathrm{PLD},i}$ rise or fall along $p_i \propto \sigma$? We resolve it first in the regime where DP-SGD operates—small sampling rates, where the advantage admits a closed form—and then extend it to arbitrary sampling rates.

Corollary 8 (Externality on $A_{\mathrm{PLD},i}$, small-p regime)**.** *Under the hypotheses of Lemma 7, assume additionally $p_g \le p_\star$ for every group, with $p_\star$ small enough that the $O(p^2)$ remainder of the subsampling expansion [16] is dominated by the leading term. Then $A_{\mathrm{PLD},i}(p_i, \sigma) = I\, p_i \,\mathrm{Adv}_G(\sigma) + O\big((Ip_i)^2\big)$, where $\mathrm{Adv}_G(\sigma) = 2\Phi(1/(2\sigma)) - 1$, with Φ the standard-normal CDF, is the advantage* (5) *of the unsubsampled Gaussian mechanism at noise multiplier σ, and*

$$\frac{\partial A_{\mathrm{PLD},i}}{\partial \varepsilon_j} \;=\; \frac{\partial \sigma}{\partial \varepsilon_j} \cdot I\, p_i \cdot \Big[\, \tfrac{\mathrm{Adv}_G(\sigma)}{\sigma} + \mathrm{Adv}_G'(\sigma) \,\Big] + O\big((Ip_i)^2\big) \;<\; 0 \tag{13}$$

for every $i \neq j$ on the small-p portion of the feasible region: tightening any other player's budget strictly raises $A_{\mathrm{PLD},i}$.

Proof idea. The single-step Poisson-subsampled trade-off function is exactly the mixture $T_{p,\sigma} = (1-p)(1-\alpha) + p\,T_\sigma(\alpha)$, so its per-step sup-gap is $p\,\mathrm{Adv}_G(\sigma)$, which composes to $Ip_i\,\mathrm{Adv}_G(\sigma) + O\big((Ip_i)^2\big)$. Along $p_i \propto \sigma$ the advantage is therefore proportional to $\sigma\,\mathrm{Adv}_G(\sigma)$, and the bracket in (13) equals $\frac{1}{\sigma}\frac{d}{d\sigma}[\sigma \mathrm{Adv}_G(\sigma)]$, which is strictly positive (Section B). Hence $A_{\mathrm{PLD},i}$ is strictly increasing along the line, and tightening ε_j raises σ and therefore $A_{\mathrm{PLD},i}$: the additional sampling outweighs the additional noise at every σ. □

Beyond the small-p regime this expansion no longer determines the sign, but the line $p_i \propto \sigma$ still constrains what a tightened ε_j can do to player i. Along the line the mechanism moves, yet its leading-order (Rényi) privacy content stays constant: the (ε_i, δ) guarantee is maintained exactly, so the contract cannot see the change. Everything happens in the higher-order shape of the privacy profile, where the effect consumes slack instead of violating the guarantee. A sign carried by higher-order terms admits no closed-form proof. Proposition 18 (Section B-C) therefore proves the invariance and certifies $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j < 0$ numerically on *every* feasible

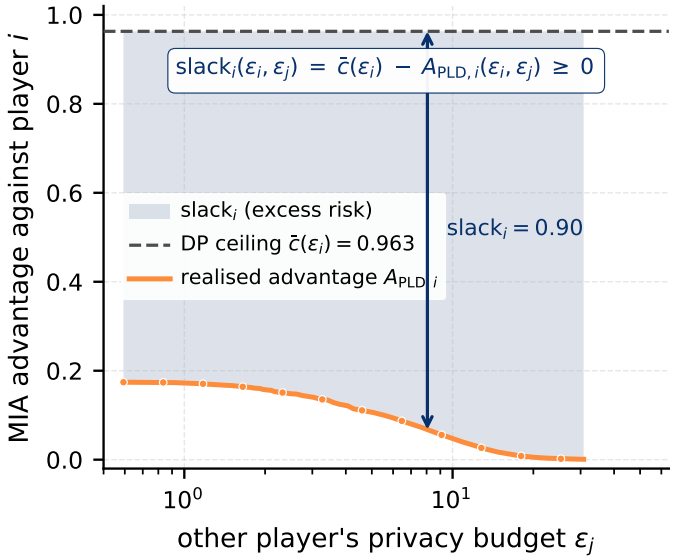


Fig. 2: **Promise versus reality at a fixed contract** (anchor $\varepsilon_i = 4$, $K = 200$). Dashed: the level-1 ceiling $A_{\varepsilon_i,\delta} = 0.963$ of the hierarchy (6). Solid: the level-2 advantage $A_{\mathrm{PLD},i}$ as the *other* player's budget ε_j varies at fixed ε_i. Shaded: the slack—the gap between the two levels. The level-1 guarantee upper-bounds the risk at every profile, but where the level-2 risk sits under it is decided by the other player's choice.

interior cell of the mechanism grid (Section D)—25,612 of 25,612 unit-stride differences, in both the cross and the own direction.

In the decomposition (9) this reads: player i's contract fixes the ceiling $A_{\varepsilon_i,\delta}$, and the other players decide where below the ceiling the realised advantage sits. So far this effect had only been observed empirically [5]. With a proven sign it becomes predictable, and a predictable advantage is what allows an informed game.

Promise versus reality at a fixed contract. The externality is visible on a single contract, with all strategy stripped away. Fix player i's budget at $\varepsilon_i = 4$ and sweep only the *other* player's budget (Fig. 2): player i's contract, and with it their line $p_i \propto \sigma$, never moves; every change below is the interaction moving them along it. The contracted vulnerability $A_{\varepsilon_i,\delta} = 0.963$ stands still, and $A_{\mathrm{PLD},i}$ stays below it everywhere—the guarantee holds—but the slack between the two swings from 0.79 to 0.96, smoothly and monotonically, exactly as Proposition 18 predicts. The (ε_i, δ) pair alone reveals none of this variation; it is the room the strategic play of Sections IV-B and IV-C exploits, and the quantity a Δ-augmented contract caps (Section VII). The full externality-response sweep, and the exhaustive sign certificate, are in Section D.

B. No symmetric equilibrium

The externality's sign says how one player's choice moves the other's slack (9); it does not say where play comes to rest. Equal protection is feasible: on every diagonal profile of the symmetric game the two slacks coincide. Do the players settle there? At exactly one budget, yes; everywhere else, no.

Proposition 9 (Diagonal trough outside utility saturation)**.** *Consider the two-player PEG with $\lambda_1 = \lambda_2 = \lambda > 0$ and $|G_1| = |G_2|$, in the small-p regime of Corollary 8. Assume $V \in C^2$ with $V' < 0$, and that the shared utility is not yet saturated at the diagonal operating point:*

$$\sigma^2 V''(\sigma) \;+\; 2\,\sigma\, V'(\sigma) \;\geq\; 0, \tag{14}$$

which for the canonical $V(\sigma) = 1/(1+\sigma^2)$ holds exactly on $\sigma \geq \sqrt{3}$. Then no interior diagonal profile $(\varepsilon^\star, \varepsilon^\star)$ with strict feasibility is a pure Nash equilibrium.

Proof idea. In the root-budget variable $s_i := \sqrt{\rho(\varepsilon_i, \delta)}$, identity (12) makes the mechanism an aggregator: $\sigma = C/(s_1 + s_2)$ and $p_i = C'\, s_i/(s_1 + s_2)$ for positive constants C, C'—each player's sampling rate is their share of the aggregate. On the diagonal the two first-order conditions coincide, and two exact identities decide the second-order test (Section B-D): the difference $\partial^2_{s_1} u_1 - \partial_{s_1}\partial_{s_2} u_1$ isolates the share term, the sum assembles the full diagonal curvature

$$4s^2\,\frac{\partial^2 u_1}{\partial s_1^2} \;=\; \left[\sigma^2 V'' + 2\sigma V'\right] \;+\; \lambda \cdot (\text{own-risk curvature}), \tag{15}$$

and in the small-p regime the own-risk curvature is strictly positive (a two-line Gaussian computation). Under (14) both terms are non-negative and the first-order candidate is a strict local *minimum* of each player's own utility: deviating in either direction pays. Diagonal points that are not critical fail the first-order test outright. Full proof in Section B-D. □

Where saturation wins: one symmetric equilibrium. The saturation condition is not decoration. V is bounded, so joint loosening buys less and less; below $\sigma = \sqrt{3}$ the shared-utility curvature in (15) turns negative, and where it outweighs the risk curvature the trough flips to a peak. The canonical instance realises the flip: the operating noise satisfies $\sigma < \sqrt{3}$ at every diagonal first-order candidate, the measured curvature is negative on a window around $\varepsilon \approx 3$, and inside it the game has *exactly one* symmetric pure equilibrium, at $\varepsilon^\star = 2.96$—confirmed on a 25-point local refinement of the action grid, every deviation losing (Section B-D). Everywhere else the diagonal remains a trough: the median best-deviation gain across the 200 diagonal cells is 0.122 utility, and alternating best-response dynamics started on the diagonal leave it at the first response in 197 of 200 runs—the other three start inside the window and stay—settling after a median of two steps at an off-diagonal pure equilibrium (Section B-E).

To see the proposition at work, play the game from any starting profile. A player asks one question: does the last unit of budget buy more model utility than it costs in weighted own advantage? The balance point is the interior best response,

$$\frac{1}{\lambda_i} \;=\; \left.\frac{\partial A_{\mathrm{PLD},i}/\partial \varepsilon_i}{V'(\sigma)\,\partial\sigma/\partial\varepsilon_i}\right|_{\boldsymbol{\varepsilon}}, \tag{16}$$

and the direction of the imbalance is the direction of the move: while the marginal utility exceeds the weighted marginal advantage, loosen; once it falls short, tighten. From here the walk to an equilibrium has four steps.

1) **On the diagonal, both directions pay—outside the saturation window.** Proposition 9 puts every unsaturated diagonal candidate at the bottom of a trough of each player's own utility, so the question above has the worst possible answer—both. On the $K = 200$ cache the trough is the rule: the median best-deviation gain across the 200 diagonal cells is 0.122 utility. Someone moves—unless play starts inside the window around $\varepsilon^\star = 2.96$, where nothing pays and the symmetric equilibrium holds.
2) **One loosening, two ledgers.** Suppose player 1 loosens; everything that follows is the aggregator (12). For player 1 the move is a new contract: a steeper line $p_1 \propto \sigma$ and a larger share of the fixed batch. The noise falls, so V rises for both players, and player 1 is sampled more under that thinner noise, so $A_{\mathrm{PLD},1}$ rises—player 1 buys the shared utility with their own exposure. Player 2 signs nothing and does nothing, yet gains twice: their line is unchanged, the falling σ slides them down it, and along the line less sampling beats less noise (Corollary 8)—$A_{\mathrm{PLD},2}$ falls while V rises.
3) **The counter-move is to tighten.** The model improvement player 2 would have loosened for has already been delivered, so the utility side of (16) has collapsed while the advantage side has not: budgets are strategic substitutes, and player 2's best response is to tighten—shed exposure, and consume what player 1 funded.
4) **Play settles at a crossing, generically off the diagonal.** By step 3, near the diagonal each best response falls in the other's budget, and an equilibrium is a crossing of the two mirror-image loci (Fig. 4). Outside the saturation window the diagonal threads the gap between them: no symmetric profile is a mutual best response, so alternating best responses leave the diagonal at the first move—in 197 of 200 runs started there—and settle, in a median of two steps, where the loci cross: one player loose and exposed, one tight and protected. The three exceptions start inside the window and stay at the symmetric equilibrium (Section B-E).

Unlike the externality results, this subsection needs more of V than its sign: the saturation curvature (14) is load-bearing, and it is exactly where it fails that symmetric play survives.

Enumeration agrees: one symmetric equilibrium, all others asymmetric. We enumerate the cleanest instance by brute force: two players, identical group sizes, identical weights, both budgets swept over the full grid. The $K = 200$ payoff matrix has 43 pure Nash equilibria: the symmetric equilibrium of the saturation window, and 42 off-diagonal equilibria in mirror pairs (Fig. 4; representative equilibria in Table I). The asymmetric equilibria are *frontier* equilibria: at all 42, the tight player sits within one grid cell of the participation floor against the loose player's budget—the same feasible infimum the coalition best response of Section IV-E selects—so the budget ratios concentrate in the narrow band the frontier's geometry dictates, $\max(\varepsilon_1^*, \varepsilon_2^*)/\min(\varepsilon_1^*, \varepsilon_2^*) \in [6.38, 6.70]$ with median 6.54. Stable asymmetric play means one player roughly $6.5\times$ looser, and the PLD-advantages split accordingly—at the most asymmetric equilibria one player's realised A reaches 0.32 while the other's stays below 0.001. Figure 3 shows the incentive at work: at any diagonal profile outside the window, lowering the own budget buys a large drop in own advantage at a modest utility cost.

The BR loci in Fig. 4 are the zero set of (16), read off the cached payoff by numeric differentiation along each player's own axis: neither side of (16) has a closed form, but the sign of their difference is measurable in every cell. The corner branches are not interior optima: against a sufficiently loose opponent, the best response is pinned to the participation frontier (Assumption 4)—the same feasible infimum the coalition best response of Section IV-E selects—and against a sufficiently tight one, to the frontier's mirror. The frontier itself is analytic. On it, player 2's rate sits at the floor $\bar{p}$, so the batch constraint pins player 1's rate at $B/|G_1| - \bar{p}$; the noise $\bar{\sigma}(\varepsilon_1)$ then solves player 1's budget alone, and the frontier is the curve on which $(\bar{\sigma}(\varepsilon_1), \bar{p})$ exhausts exactly ε_2. So on two of its three branches the best response is a closed-form curve, and the equilibria inherit that structure: the tight player's coordinate is exact, the loose player's is a near-indifferent position along the wall—the loose payoff varies by less than 10^{-4} across the highlighted segment, so the 42 grid equilibria are best read as samples from a one-parameter family of frontier equilibria. Grid resolution moves the equilibrium *count*, not the structure: at $K = 22$ the same game resolves 7 pure NEs, at $K = 200$ it resolves 43—the stepped BR loci of a finer grid cross more often. What is invariant, and is the claim we stake, is the structure: mirror pairs along the crossing band, plus the single symmetric equilibrium of the saturation window, already present at $K = 22$; we therefore quote $K = 200$ throughout and read individual NE coordinates as samples from the crossing band. The common privacy weight moves the picture only at its extremes: $\lambda \in \{0.125, 0.25, 0.5, 1\}$ yields 34–62 pure NEs of the same structure, but at $\lambda = 2$ the grid has *no* pure equilibrium at all—only mixed play survives—and at $\lambda = 4$ a single corner equilibrium $(0.5, 0.5)$ remains (Section E).

What survives of symmetry. Two things, of very different character. Inside the saturation window, symmetry survives outright: the pure equilibrium at $\varepsilon^\star = 2.96$ gives both players the same budget and the same realised advantage (0.104 each), and it is the equilibrium with the least aggregate advantage of the entire set (Section V-C). Outside the window, symmetry survives only as a lottery. Restricted to the two budgets of a mirror pair—take $(1.12, 7.13)$—the PEG is a Hawk–Dove game [6]: each player prefers the tight, protected role ($u = 0.50$, against 0.25 for the exposed role), both symmetric profiles of the restriction are unstable, and the restriction's only symmetric equilibrium is mixed [47], playing tight with probability 0.45 at

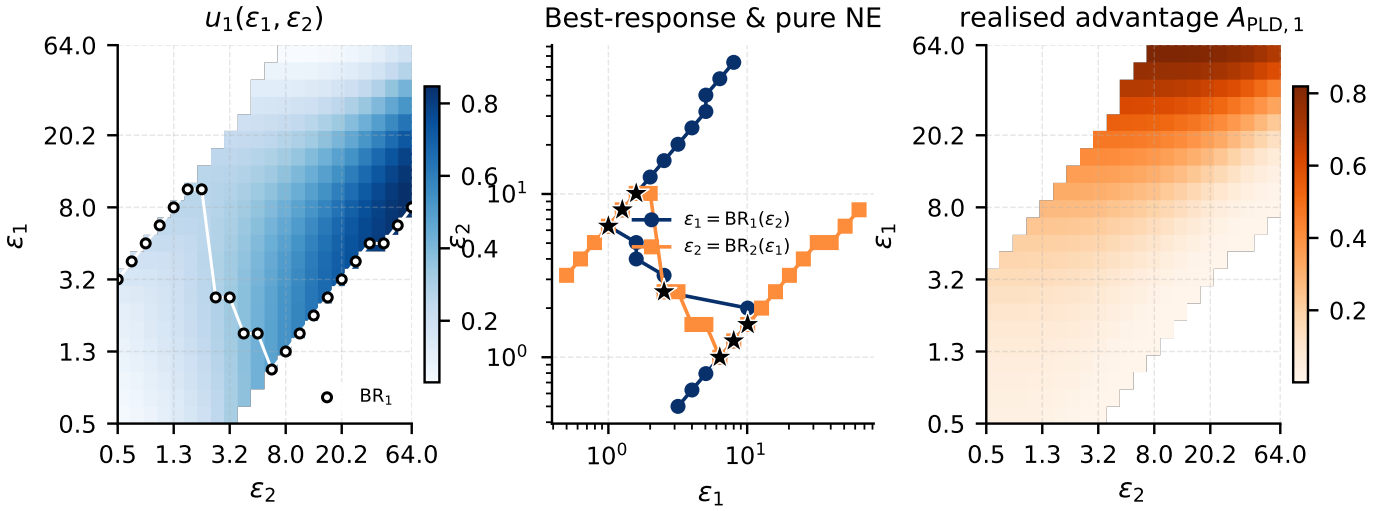


Fig. 3: **Anatomy of the symmetric 2-player PEG** ($K = 22$ shown for legibility). Payoff surface $u_1(\varepsilon_1, \varepsilon_2)$ (left), best-response correspondences with pure NEs as black stars (centre), and realised MIA advantage $A_{\mathrm{PLD},1}$ (right). Every pure NE lies off the diagonal.

TABLE I: Representative pure Nash equilibria of the symmetric 2-player PEG on the $K = 200$ log-spaced action grid ($\varepsilon \in [0.5, 64]$, $|G_1| = |G_2| = 5000$, $B = 128$, 5 epochs, $\delta = 10^{-12}$, $\lambda_1 = \lambda_2 = 1$). One member of each mirror pair is shown; the full set has 43 equilibria: 42 off-diagonal in mirror pairs and the symmetric equilibrium of the saturation window. PoEV (Definition 14) is measured against the uniform baseline $\varepsilon^{\mathrm{uni}} = 5.73$ with $\sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\mathrm{uni}}) = 0.310$. The social optimum $(3.35, 24.13)$ is not an equilibrium; the best equilibrium is the symmetric one.

	$(\varepsilon_1^*, \varepsilon_2^*)$	$A_{\mathrm{PLD},1}$	$A_{\mathrm{PLD},2}$	$\sum_i A_{\mathrm{PLD},i}$	PoEV
symmetric (window)	$(2.96, 2.96)$	0.104	0.104	0.207	-0.33
median	$(1.12, 7.13)$	0.001	0.252	0.253	-0.18
most inflationary	$(1.61, 10.79)$	0.001	0.325	0.325	$+0.05$

value 0.35—strictly between the two role payoffs (Section B-E). That lottery is fair, but every draw of it hands one player the exposed role. Any asymmetry between the players breaks the tie and selects the convention; preference asymmetry is the one our players have, and we turn to it next.

C. Whose slack shrinks: self-sorting

Under exact symmetry the equilibria come in mirror pairs, and which player concedes is settled only by the lottery over roles of Section IV-B. Any preference asymmetry breaks the tie, and it breaks it in a fixed direction.

Proposition 10 (Self-sorting, 2-player PEG)**.** *Consider the 2-player PEG with $\lambda_1 > \lambda_2 > 0$ (player 1 the more privacy-sensitive), $|G_1| = |G_2|$, V strictly decreasing, own-budget monotonicity $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_i > 0$, and the exchange condition (S) of Section B-F. Then every pure Nash equilibrium $(\varepsilon_1^\star, \varepsilon_2^\star)$ satisfies $\varepsilon_1^\star \le \varepsilon_2^\star$.*

Proof idea. Suppose the *more* privacy-sensitive player held the higher budget. At equilibrium neither player can profit by swapping to the other's budget; dividing the two resulting no-swap inequalities and applying condition (S) forces $\lambda_2 \ge \lambda_1$—a contradiction. Full proof in Section B-F. □

Condition (S) holds throughout the high-noise regime and is verified numerically at every equilibrium candidate we report (Section B-F).

The sweep confirms the sorting. Fixing $\lambda_2 = 1$ and sweeping $\lambda_1 \in \{0.25, 0.5, 1, 2, 4\}$ yields 37–61 pure NEs per ratio (Fig. 6). The NE cloud migrates exactly as Proposition 10 predicts: when player 1 values privacy less, the equilibria sit in the region where player 1 holds the high budget and the higher realised advantage; as λ_1/λ_2 crosses unity, the cloud migrates through the symmetric configuration into the mirror region. Sorting compliance—the fraction of equilibria with the higher-λ player on the lower budget—is 170 of 174 across the sweep, with all four exceptions at the mildest ratio $\lambda_1/\lambda_2 = 0.5$. Preference is not the only asymmetry that assigns the roles. Sweeping the group-size ratio $r = |G_1|/|G_2|$ across two orders of magnitude (all else symmetric), the smaller group plays the smaller equilibrium budget and ends at the smaller realised advantage at every ratio (Fig. 5): with player 1 in the minority ($r < 1$), equilibria have $\varepsilon_1^* \in [1.5, 3]$ against $\varepsilon_2^* \approx 10$, yielding $A^*_{\mathrm{PLD},1} < 0.01$ against $A^*_{\mathrm{PLD},2} \in [0.24, 0.32]$, and the pattern mirrors for $r > 1$. The mechanism is legible in (4): a small group demanding strong privacy contributes little to the expected batch, the majority's sampling rate rises to compensate, and its realised advantage rises with it. The minority free-rides on the shared mechanism [28]—a privacy analogue of the classical result that small groups benefit disproportionately from a common resource.

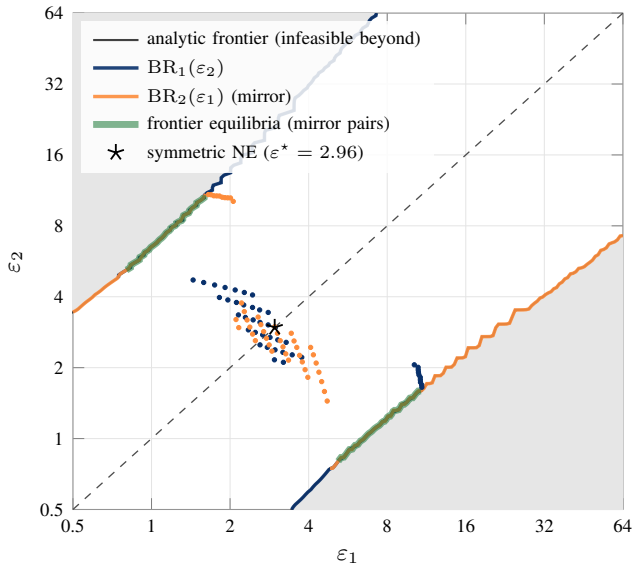


Fig. 4: **The best-response landscape** ($K = 200$). Each player's best response has three branches, read off the zero set of the measured first-order condition. Against a tight opponent (lower/left leg) the best response runs along the *upper* feasibility wall: loosening pays all the way to the frontier. Against a loose opponent (upper/right leg) it runs along the *lower* wall: tightening pays all the way to the participation floor. In between, an interior branch crosses the diagonal (dashed); it is drawn as points because the payoff there is flat to within 10^{-2} of its maximum and the ridge position is near-indifferent. Grey marks the infeasible profiles; the dark curve is the analytic frontier—the floor rate $\bar{p}$ for one player, the batch pinning the other. The equilibria sit where the mirror-image branches meet: the mirror pairs occupy the highlighted *frontier segments*—the tight player pinned at the feasible infimum, the loose player near-indifferent along the wall, budget ratios $6.38\times$–$6.70\times$ (median $6.54\times$)—and the saturation window contributes the single symmetric equilibrium at $\varepsilon^\star = 2.96$ (star), where the interior branches cross.

The direction is fixed: the *more* privacy-sensitive player ends at the lower budget *and* the lower realised advantage; the slack that shrinks belongs to the player who cares least. The exposed player is not exploited—at equilibrium the shared model utility compensates them. But the outcome is one the iDP contract never priced: your realised protection is set by your neighbours' preferences, not your own.

D. Existence, and why enumeration is necessary

Symmetry breaking says where equilibria are not; it does not yet show that any exist. In the low-sampling region where DP-SGD operates, the same substitutes structure that pushes the players apart also delivers existence.

Theorem 11 (Pure NE existence on a substitutes box, $n = 2$)**.** *Let $\mathcal{B} = [\varepsilon_{\mathrm{lo}}, \varepsilon_{\mathrm{hi}}]^2$ be a box of feasible profiles on which $p_g \le p_\star$ (as in Corollary 8) and the budgets are strategic substitutes, $\partial^2 u_i / \partial\varepsilon_1 \partial\varepsilon_2 \le 0$. Then the PEG restricted to $\mathcal{B}$ has a pure Nash equilibrium [48], [49], and any such equilibrium whose best responses over the full action set remain in $\mathcal{B}$ is an equilibrium of the PEG itself.*

Both hypotheses hold in the practically relevant regime: DP-SGD sampling rates sit at 10^{-4} to 10^{-2}, placing operation in the small-p regime of Corollary 8, and there the substitutes condition reduces, for the canonical V, to the curvature bound $\sigma \le \sqrt{3}$, which holds on the less-noisy portion of the caches' operating range $\sigma \in [0.5, 5]$ (proof and the lattice structure of the equilibrium set in Section B).

Two limitations remain. Existence on the full feasible region, including the saturation corner $p_g \to 1$, is open; there we rely on exhaustive grid enumeration (Section IV-B). And this order-theoretic (lattice) route is intrinsically two-player: substitutes cannot be re-ordered into complements for every pair at once when $n > 2$. The aggregative structure of (12)—opponents enter only through a sum—is the natural route to larger games, in the Cournot tradition of [50]; we leave that route open and use enumeration and fictitious play at $n > 2$ (Section V-A).

Enumeration is a necessity, not a convenience: the familiar shortcut—a potential function whose minimiser is automatically an equilibrium—is unavailable in principle.

Proposition 12 (The PEG is not a potential game)**.** *Under* (7) *and the mechanism of Definition 1, the PEG admits no exact, weighted, or ordinal potential in the sense of [31].*

The Price-of-Privacy game was a potential game precisely because its privacy cost was separable [30]; the mechanism's coupling is what destroys the structure (proof in Section B). This is why our existence route trades the concavity that the classical existence arguments require for the monotonicity that strategic substitutes provide, and why the sign analysis of Corollary 8 and Proposition 18 is the load-bearing input.

Where V enters as more than a sign. The externality proofs never touch V, the best-response condition (16) uses only $\operatorname{sgn} V'$, and the coalition corollary of Section IV-E uses only the externality sign on the target's coordinate; Proposition 20 (Section C) records this formally. The one exception is deliberate: the symmetry-breaking result of Section IV-B reads V's saturation curvature (14), and the location of the symmetric equilibrium moves with the shape of V (Section C-A). The experimental choice $V(\sigma) = 1/(1+\sigma^2)$ is a representative of a class for every sign-driven result, and the explicit curvature condition states exactly what the exception needs.

E. Coalitions: the attack is a corollary

Every player so far weighs only their own harm. We now let a subset commit to a joint action first, with the remaining players responding—a Stackelberg game with malice weight κ on the target's vulnerability (Definition 5)—and read off the coalition's best response in the high-malice limit.

Corollary 13 (Coalition best response recovers the attack)**.** *Fix a target t, a coalition $C \subseteq N \setminus \{t\}$ of size k, and any*

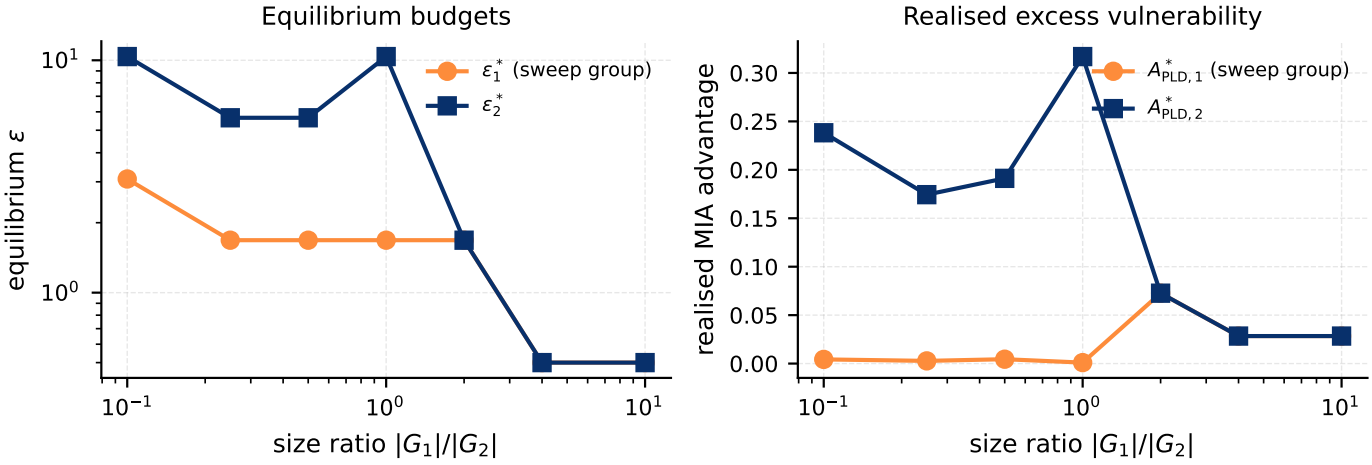


Fig. 5: **Group-size asymmetry.** Equilibrium budgets (left) and realised MIA advantages (right) as a function of the size ratio $|G_1|/|G_2|$. The minority group plays the lower budget and ends at the lower realised advantage at every ratio; the flip at $r{=}1$ is where the minority/majority labels exchange.

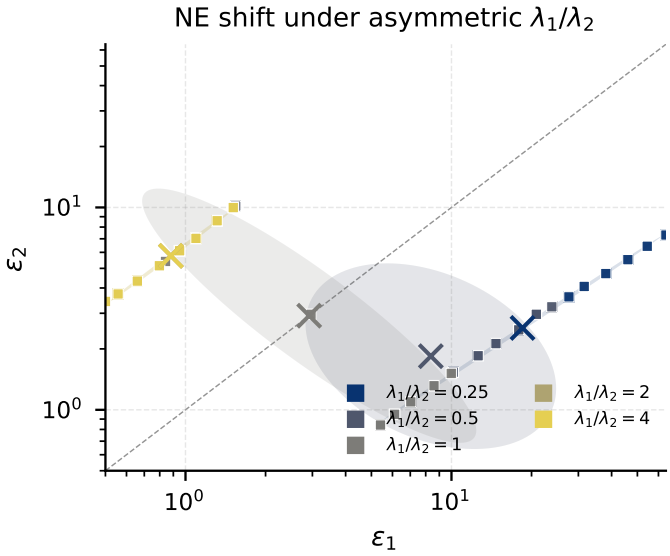


Fig. 6: **Preference asymmetry** ($K = 200$, $\lambda_2 = 1$). Pure NE clouds under $\lambda_1/\lambda_2 \in \{0.25, 0.5, 1, 2, 4\}$ (centroids and 68% dispersion ellipses). As player 1 becomes more privacy-sensitive the cloud migrates from lower-right (player 1 exposed) to upper-left (player 2 exposed), never crossing the diagonal: heterogeneity self-sorts instead of averaging out.

actions ε_{-C} for the others. As $\kappa \to \infty$ in Definition 5, the coalition's Stackelberg best response is coalition-symmetric—every colluder plays the same budget—at the participation-preserving minimum *$\varepsilon_k^\star(\varepsilon_{-C})$, and the target advantage $A_{\mathrm{PLD},t}$ induced at that best response is non-decreasing in k.*

Proof idea. $\partial A_{\mathrm{PLD},t}/\partial \varepsilon_i < 0$ for every $i \in C$ (Proposition 18; analytic at small p, certified numerically beyond), so the coalition pushes every budget as low as feasibility admits. Below the participation floor a colluder's sampling rate reaches zero (Assumption 4), so they drop out of (4) and stop raising σ—pushing lower only forfeits leverage; the optimum is therefore the *feasible infimum*, not the grid minimum. Adding a member enlarges the set of budgets pulling σ up, so $A_{\mathrm{PLD},t}$ is non-decreasing in k. □

This is the budget-manipulation attack of [5], recovered as rational play: any coordinated subset of data holders whose joint utility weights the target's harm arrives at the same action vector, whether or not anyone calls it an attack. The equivalence is all the corollary claims—the empirical demonstration is prior work's—and Section VI shows the attack's force is contingent on a target that does not respond.

V. Numerical Evaluation

The analysis of Section IV settled the two-player game with its evidence in place. What remains are the questions that leave that setting, each a subsection: whether the picture survives more players (Section V-A); whether it is the mechanism or the paradigm (Section V-B); and what equilibrium play costs in aggregate (Section V-C). Section V-D closes by asking whether any of it survives a *measured*, rather than analytical, adversary—the answer draws the practical boundary of the phenomenon. Sections V-A to V-C use the pipeline of Section IV; Section V-D instead trains real DP-SGD shadow models and measures the advantage with LiRA [20]. Supporting sweeps live in the appendices: common-weight shifts (Section E), the raw externality response and its finite-difference sign certificate (Section D), model-utility ablations (Section C-A), and grid and threshold sensitivity (Section H).

A. Does the asymmetry survive more than two players?

We sweep $n \in \{3, \ldots, 10\}$ on a coarse $K = 7$ grid, enumerating pure NEs exhaustively for $n \le 5$ and running 32 seeds of Monte-Carlo fictitious play for $n \ge 6$ (Fig. 7). Under heterogeneous weights $\lambda_i \in \mathrm{geomspace}(0.25, 4)$, the

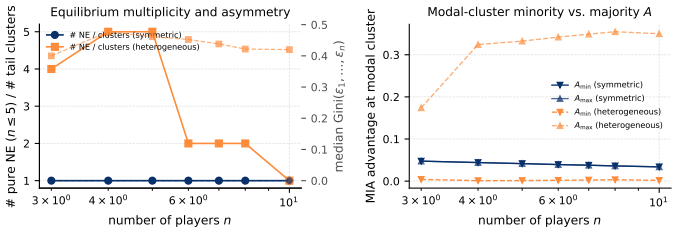


Fig. 7: n**-player PEG** on the K=7 grid: brute-force NE for $n \leq 5$, fictitious-play tails for $n \geq 6$ (32 seeds $\times$ 400 rounds). *Left axis:* fraction of observations whose budget order is the exact reverse of the weight order (self-sorting compliance) under heterogeneous λ_i. *Right axis (log):* the $A_{\max}/A_{\min}$ advantage ratio across observed profiles. Sorting is exact for $n \leq 8$ and near-exact at $n = 10$; the advantage asymmetry persists at every n.

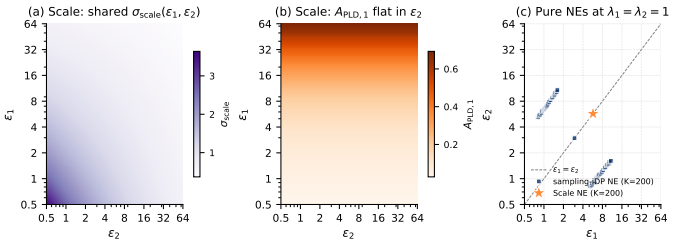


Fig. 8: **Sampling-based versus sensitivity-based iDP, same game** ($K = 200$, $\lambda = 1$). The sensitivity-based realisation (single orange star) has exactly one pure NE, on the diagonal at $(5.73, 5.73)$; the sampling-based realisation (blue) has 48 pure NEs, all off-diagonal. The strategic asymmetry is a property of the budget-coordination mechanism, not of individual budgets.

self-sorting signature is exact at every $n \in \{3, \ldots, 8\}$—the rank order of equilibrium budgets is the reverse of the rank order of weights in 100% of observations—and 83% at $n = 10$, where the fictitious-play tail occasionally swaps adjacent, nearly-tied weights. Realised advantages remain strongly asymmetric throughout: the extreme players' advantage ratio $A_{\max}/A_{\min}$ runs from roughly 480 at $n = 3$ to 75 at $n = 10$. Under fully symmetric weights, fictitious play on this coarse grid settles onto symmetric profiles for $n \leq 8$; this is the expected discretisation effect—the diagonal deviation gain of Proposition 9 is second-order and falls below a $K = 7$ grid step—and at $n = 10$ the dynamics again find asymmetric profiles with advantage ratios near 5. Resolving the symmetric-n case on a fine grid is a computational, not conceptual, gap; we flag it as open.

B. Is it the mechanism or the paradigm?

The asymmetry could conceivably be a feature of individual budgets per se. It is not: it is a feature of *how sampling-based iDP reconciles them*. We replay the identical game—same groups, weights, grid, utility—through the *sensitivity-based* iDP realisation of [3], in which each group receives a personal clipping norm under a shared noise multiplier and each record's noise-to-sensitivity ratio is preserved by construction, so $A_{\mathrm{PLD},i}$ depends on ε_i alone and the players interact only through the shared $V(\sigma)$.

The contrast is total (Fig. 8). At $\lambda = 1$ and $K = 200$, the sensitivity-based game has *exactly one* pure Nash equilibrium, and it is symmetric: $(\varepsilon_1, \varepsilon_2) = (5.73, 5.73)$. The sampling-based game on the identical grid has 43, all but the single saturation-window equilibrium asymmetric. The same uniqueness-and-symmetry holds across the common weight sweep ($\lambda \in \{0.25, 0.5, 1\}$ each yield a single diagonal NE). This is the cleanest statement of the paper's scope: individual privacy budgets do not by themselves create the strategic asymmetry; the joint (σ, p_g) solve of the sampling-based realisation does. It is also, in effect, a control experiment for the theory— remove the coupling that Lemma 7 formalises, and the Hawk–Dove structure disappears.

C. How costly is equilibrium play?

Vulnerability is not conserved: equilibria that drive the players' realised advantages apart may still lower—or raise—their sum. We measure which with a value-side analogue of the price of anarchy [27].

Definition 14 (Price of externality-vulnerability)**.** Given a symmetric baseline profile $\boldsymbol{\varepsilon}^{\mathrm{uni}}$ and a reference profile $\boldsymbol{\varepsilon}^{\star}$,

$$\mathrm{PoEV}(\boldsymbol{\varepsilon}^{\star}) := \frac{\sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\star}) - \sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\mathrm{uni}})}{\sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\mathrm{uni}})}.$$

Against the geometric-midpoint baseline $\varepsilon^{\mathrm{uni}} = 5.73$ (aggregate advantage 0.310), the 43 equilibria of the canonical game span $\mathrm{PoEV} \in [-0.33, +0.05]$ with median -0.18 (Table I): most equilibria *deflate* aggregate advantage relative to uniform play, because the utility term pulls both players toward moderate noise. Three readings temper that comfort. First, the aggregate hides the asymmetry: away from the saturation window, the median equilibrium assigns one player more than two orders of magnitude the advantage of the other (Table I). Second, the spread across equilibria is itself the problem—the best equilibrium is the symmetric one ($\mathrm{PoEV} = -0.33$), the social optimum $(3.35, 24.13)$ is not an equilibrium at all, the worst equilibrium inflates aggregate advantage over the baseline ($+0.05$), and nothing in the mechanism selects among them. Third, the sign flips with the preference regime (Section E): at utility-heavy weights ($\lambda \leq 0.25$), *every* equilibrium is inflationary, with PoEV up to $+1.65$—rational play then raises aggregate vulnerability by up to 165% over uniform budgets. Aggregate welfare at equilibrium is, in short, unmanaged: sometimes better than uniform, sometimes much worse, and never chosen.

D. Does the game survive a measured adversary?

Everything so far prices risk at the middle level of the hierarchy (6)—the mechanism-specific, PLD-exact bound on what any adversary can extract from the deployed mechanism. A fair question is whether the strategic structure survives one level further down, when the advantage is *measured* instead. We instantiate the two-player game end-to-end on Adult: for

each profile of a 5×5 budget grid ($\varepsilon \in \{2, 4, 8, 16, 32\}$, 5000 records per silo) we train 128 DP-SGD shadow models at the mechanism solution (σ, p_1, p_2), score per-record membership with LiRA [20], and rebuild the payoff matrix from the measured advantage and measured test accuracy (Fig. 9).

The measured game degenerates—informatively. On this workload the LiRA advantage is flat at the measurement floor across the entire grid ($A^{\mathrm{LiRA}} \in [0.107, 0.115]$, per-silo AUC ≤ 0.506) while the analytical bound spans $[0.004, 0.870]$: the attacker cannot resolve the leakage the accountant prices, the externality term of (7) is thus constant in the players' actions, and the equilibria slide to the high-ε corner where accuracy dominates ($(16, 16)$ and $(32, 32)$). The analytical game on the same coarse grid puts its equilibrium at the *opposite*, low-budget corner $(2, 2)$—a 5×5 grid is too coarse to resolve the off-diagonal band of Section IV-B, but the corner contrast is the point: the measured incentives are not a perturbation of the analytical ones, they are advantage-blind. This is the second inequality of the hierarchy (6) at its loosest—on the $(16, 16)$ profile the pooled attack ROC sits at chance ($A_{\mathrm{LiRA}} = 0.010$) against an optimal-adversary $A_{\mathrm{PLD}} = 0.58$, and the flat $[0.107, 0.115]$ band of the per-record statistic is that estimator's finite-sample floor at 128 shadow models—and it is the privacy-onion effect of [24] made strategic: *the game's asymmetric structure stands or falls with the attacker's ability to realise the externality*. Two consequences follow. For the theory's scope: the equilibrium asymmetry binds in regimes where realised attack power tracks the accountant—strong attackers, vulnerable records, or the worst-case adversary the contract itself must assume. For practice: a deployment whose realistic adversary sits at the measurement floor will see no strategic pressure on budgets—but a risk assessment cannot be written against today's measurement floor, and the level-2 game prices exactly the most any adversary can ever extract from the deployed mechanism. Closing the gap between the two—a measured instance with above-floor attack power, e.g. canary-style outliers—is the natural next experiment.

VI. From Equilibria to Attacks

The equilibrium analysis treats every player as ordinarily self-interested. This section turns the malice weight back on and works out the attack picture that Corollary 13 predicts—and its two qualifications, both of which matter for threat modelling: the attack presumes a victim who does not respond, and an attacker who plays a repeated game does not attack at the minimum budget. Every attack quantity in this section is a level-2 advantage in the sense of (6): what the coalition moves is the mechanism-specific bound on any adversary's advantage against the resulting mechanism—the attack manipulates the mechanism, not the attack tooling—so raising it raises the ceiling on every downstream attacker at once.

A. Coordination amplifies the externality against a passive target

We instantiate Corollary 13 with five symmetric players, the target pinned at $\varepsilon_t = 32$ (the oblivious-victim threat model of [5]), non-colluders passive at $\varepsilon = 32$, and a coalition of k players committing to the single budget that maximises the target's realised advantage.

The target's advantage climbs monotonically and steeply in coalition size (Fig. 10): $A_{\mathrm{PLD},t} = 0.442$ with no coalition, 0.483, 0.531, 0.607 at $k = 1, 2, 3$, and 0.728 at $k = 4$—a 65% inflation over baseline, mirroring the empirical attack success reported by [5]. The coalition's optimal budget tracks the participation-preserving minimum of Corollary 13: $\varepsilon_C \approx 6.8$ for $k \leq 3$, relaxing to $\varepsilon_C \approx 3.2$ at $k = 4$ as the collective batch constraint slackens. The strategic content is the recovery, not the curve: this is what *any* coordinated subset of data holders does if its joint utility weights the target's harm, no adversary model required.

B. The attack collapses against an aware target

Pinning the target is the right model for an oblivious victim, but it is not the game: an aware target sees the coalition's budgets and can react. We rerun the same scenario with the target as a Stackelberg follower, best-responding on a 14-point log grid to whatever the coalition commits, all else unchanged.

The attack collapses (Fig. 11). Across $k \in \{1, 2, 3, 4\}$ the aware target's realised advantage is $A^{\star}_{\mathrm{PLD},t} \in \{0.014, 0.010, 0.018, 0.041\}$—18–53$\times$ below the passive-target curve and an order of magnitude below even the no-coalition baseline. For $k \leq 3$ the leader budget stays interior or at the loose ceiling ($\varepsilon^{\star}_C \in \{44.1, 6.8, 64\}$) while the target retreats to $\varepsilon^{\star}_t \in [6.8, 14.4]$, where the mechanism-side externality on their advantage is small. At $k = 4$ the leader argmax moves to the grid floor—a *poisoning corner* at $\varepsilon_C = 0.5$ that drags the target down to $\varepsilon^{\star}_t = 0.73$—but even this corner yields only $A^{\star}_{\mathrm{PLD},t} = 0.041$ while collapsing the model utility the coalition itself trains for; the interior optimum, continuous with $k \leq 3$, does even less for the attacker (Fig. 11, hollow marker). The reading for threat modelling is sharp: the coalition attack is an attack on *unawareness*. Its empirical force in prior work comes from targets that hold their budget fixed—rational for a nominal-information player, per Remark 6—and a single round of informed best response recovers almost the entire loss. Disclosure of the mechanism's coupling to contributors is therefore not just transparency hygiene; it is quantitatively the cheapest defence available.

C. A repeated attacker learns the target—and does not play the minimum budget

Real deployments are repeated: budgets are re-declared, models re-trained, and an attacker faces a target whose privacy preference it does not know. We formalise the natural repeated extension—the *Bayesian Repeated PEG* (BR-PEG)—and extract its one qualitative surprise. Two players: an attacker M and an honest target H whose type $\theta_H = (\lambda_H, n_H)$ is drawn from a finite set Θ known to M. Each round, M commits to a budget ε^t_M; H observes it and myopically best-responds; the mechanism returns the round's outcomes, and M Bayes-updates its posterior over θ_H from H's visible response. M's score is $\sum_t A^t_H$. The game is the Stackelberg-learning template of

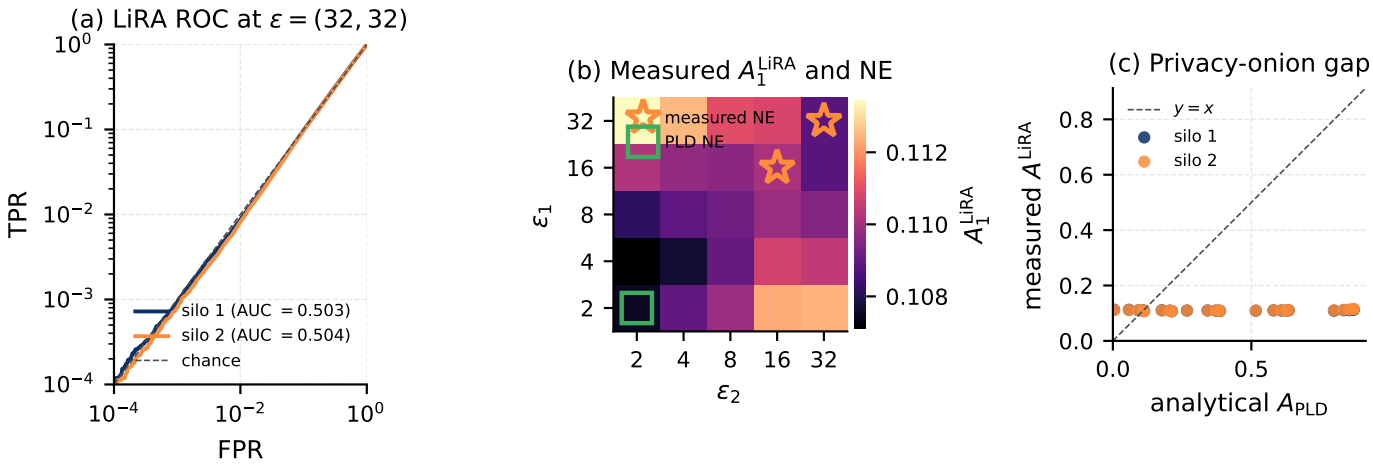


Fig. 9: **Measured instance of the 2-player PEG** (Adult, MLP, 128 LiRA shadow models per profile, 5×5 grid $\varepsilon \in \{2, \ldots, 32\}$, 5000 records/silo). *Left:* pure NE of the analytical game on the same coarse grid (low-budget corner $(2, 2)$). *Centre:* pure NE of the measured game (high-budget corner). *Right:* measured versus analytical advantage across profiles. LiRA sits at the measurement floor ($A^{\text{LiRA}} \in [0.107, 0.115]$) while the analytical bound spans $[0.004, 0.870]$; with the externality term flattened, the measured equilibria collapse to the corner where accuracy dominates. The strategic asymmetry requires attack power above the measurement floor.

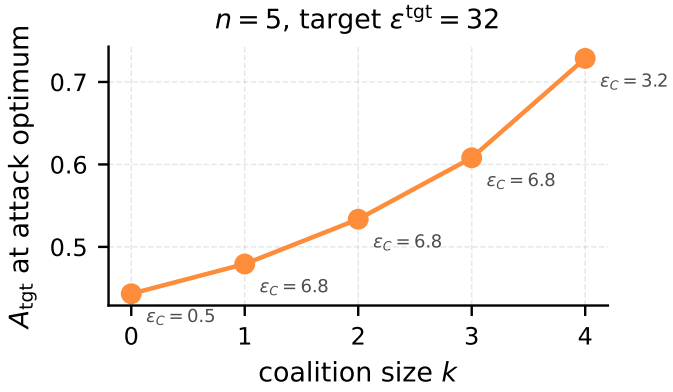


Fig. 10: **Coalition attack against a passive target** ($n = 5$, target pinned at $\varepsilon_t = 32$). Target's realised MIA advantage versus coalition size k, with the coalition playing its target-maximising joint budget (annotated). The curve realises Corollary 13: monotone in k, reaching $A_{\text{PLD},t} = 0.728$ (+65% over the k=0 baseline) at $k = 4$.

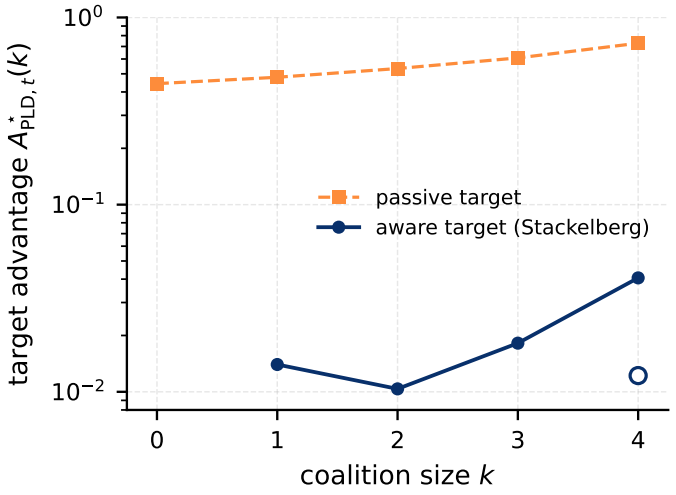


Fig. 11: **The coalition attack against an aware target** (log-y). Orange squares: passive target pinned at $\varepsilon_t = 32$ (as in Fig. 10), climbing to $A_{\text{PLD},t} = 0.729$ at $k = 4$. Blue circles: target best-responding to the coalition's commitment; $A^{\star}_{\text{PLD},t} \leq 0.041$ throughout, 18–53$\times$ below the passive curve. At $k = 4$ the filled marker is the leader argmax (a grid-floor poisoning corner that also ruins the coalition's own utility); the hollow marker is the interior optimum that a coalition with any self-interest plays.

[51]–[53] with the tabular follower payoff replaced by the DP accountant, and the myopic-informed case of repeated games with one-sided incomplete information [54]; Section F gives the full formalism.

Proposition 15 (The Bayesian-greedy attack is not min-budget)**.** *In the BR-PEG with H myopic and M posterior-greedy on a finite action grid, suppose (I) the observation likelihood identifies $\theta^{\star}_H$ on the equilibrium path [55] and (A) the elicited-advantage map $G(\varepsilon_M)$—the target's realised advantage A_H at the profile in which M plays ε_M and H plays $\varepsilon^{\text{BR}}_H(\varepsilon_M; \theta^{\star}_H)$—has a unique maximiser $\varepsilon^{\star}_M > \varepsilon^{\min}$. Then the posterior-greedy action equals $\varepsilon^{\star}_M$ for all sufficiently large t, almost surely: the learned attack is bounded away from the minimum budget.*

The proof (Section F) is a uniform-convergence argument: the posterior collapses on the true type, the posterior-averaged score converges uniformly on the finite grid, and the argmax locks onto the true-type optimum. The content is assumption

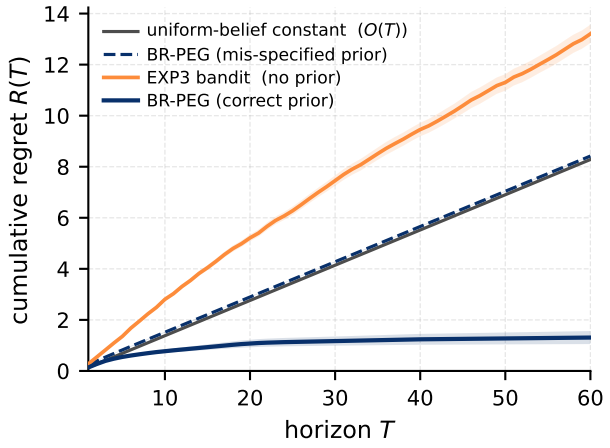


Fig. 12: **Cumulative regret of repeated attackers** against the full-information Stackelberg oracle (5 overlapping target types, noise of standard deviation 0.15 on the target's observed utility, 7-arm budget grid, mean ± 1 SE over 40 seeds). The posterior-greedy BR-PEG attacker with a correct prior plateaus ($R(60) = 1.31$); a misspecified prior and a static uniform belief pay linear regret (8.4 and 8.3); the prior-free EXP3 bandit pays 13.2. Learning the target's type costs the attacker a constant, not a rate.

(A) being *true* for the iDP mechanism, which we verify numerically: the advantage-maximising budget against a best-responding target is interior because slamming ε_M to the floor drives σ up and pushes the target into retreat, whereas a moderate ε_M keeps the target comfortable at a budget whose realised advantage the externality then inflates. In the canonical instance the oracle attack sits at $\varepsilon_M^\star \approx 3.2$ and elicits $A_H \approx 0.454$ per round—roughly twice the naive minimum-budget attack and fifty times the benign baseline (Section F).

Learning speed is the remaining question: how much does not knowing θ_H cost? Figure 12 reports cumulative regret against the full-information oracle on a harder instance designed to make identification non-trivial (five overlapping types whose best responses coincide at some arms, observation noise on H's reported utility, 40 seeds, $T = 60$). The posterior-greedy attacker with a correctly specified prior pays a bounded, front-loaded cost ($R(60) = 1.31 \pm 0.26$, flattening after roughly ten identification rounds), while both non-learning baselines pay linearly growing regret (uniform-belief 8.3, misspecified prior locked on a wrong type 8.4 at $T = 60$) and a prior-free adversarial bandit (EXP3 [56]) pays 13.2. For defenders the message is unpleasant but crisp: type uncertainty is not a durable shield—a Bayesian attacker converts it into a constant, not a rate.

VII. Mitigation: Change the Contract, Not the Players

If asymmetric vulnerability is equilibrium behaviour, mitigation cannot consist of detecting deviants—at equilibrium there are none. Two repair strategies remain: discipline the *reports* (mechanism design on the budget channel) or discipline the *contract* (bound what any profile can do to a player). The first is unavailable; the second provably works.

No report-based fix. Viewed as a direct-revelation mechanism on the budget channel, sampling-based iDP without transfers is not dominant-strategy incentive-compatible: a player's utility-maximising budget report necessarily depends on the others' declarations.

Proposition 16 (Non-strategyproofness of budget declaration)**.** *For any player i with $\lambda_i > 0$ there exist opponent declaration profiles $\hat{\varepsilon}_{-i}^{\mathrm{L}}$, $\hat{\varepsilon}_{-i}^{\mathrm{T}}$ such that player i's best-response reports against them differ. In particular, no report is a dominant strategy, and the identity mechanism on the budget channel is not DSIC.*

The proof (Section G) chains Lemma 7 through the utility gradient and exhibits a numerical witness: against a loose opponent the best report is $\hat{\varepsilon}_1 \approx 2.9$, against a tight one ≈ 1.2. The classical rescues do not apply—the large-market route of [57] needs each agent's influence on the mechanism to vanish, whereas each silo moves (4) at constant magnitude, and the truthful-elicitation results of [35]–[37] model privacy costs that vanish without influence, whereas here every report moves σ. Section G sets up the type-channel (Myerson-style) design problem this opens and why it is harder than an auction: a shared σ, interim advantages coupled through all types, and no monetary transfer.

The contract-based fix. The $(\varepsilon_i, \delta, \Delta)$-iDP contract of [5], [17] adds one clause to the individual contract: the realised mechanism must stay within Δ-divergence of a reference, equivalently the slack of (9) is capped. In the vocabulary of this paper, the clause is exactly an externality bound:

Proposition 17 (Δ-neutralisation of the externality)**.** *Restrict the action space to profiles with $\mathrm{slack}_i(\boldsymbol{\varepsilon}) \le \Delta$ for every player. Then at every feasible profile*

$$A_{\varepsilon_i,\delta} - \Delta \;\le\; A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) \;\le\; A_{\varepsilon_i,\delta},$$

and for any two feasible profiles agreeing in player i's own budget, $\left|A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) - A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}')\right| \le \Delta$.

Proof. Both claims are immediate from the identity (9) and $\mathrm{slack}_i \in [0, \Delta]$: the realised advantage is the ceiling minus the slack, and two profiles sharing ε_i share the ceiling, so their advantage difference is a difference of slacks. □

The proposition is deliberately elementary—that is its virtue. In the terms of the hierarchy (6), the Δ-clause pins the distance between levels 1 and 2: the level-2 advantage may sit at most Δ below the level-1 ceiling, so no unilateral or joint deviation by the other players, coordinated or not, can move player i's realised advantage by more than Δ at fixed ε_i. The coalition curve of Fig. 10 is capped at height Δ above its floor, and the strategic swing of Section IV-B is confined to a Δ-band around each contract. What Δ-iDP does *not* automatically deliver is a well-behaved residual game—whether an equilibrium of the restricted game exists, is unique, or is symmetric, and what

utility cost the restriction imposes through forced noise, are open (we return to this in Section VIII). The equilibrium analysis of this paper is the reason to pay that cost: without the clause, the contract's realised meaning is set by everyone else's play.

VIII. Discussion and Conclusion

Our work reframes the known fragility of sampling-based iDP: the elevated realised vulnerability that prior work attributed to adversaries is what rational play produces on its own. The Privacy Externality Game generically refuses symmetric play: wherever the shared utility is unsaturated, the diagonal is a trough of each player's own utility (proven in the small-p model, measured on the exact mechanism), deviations from it are self-reinforcing because budgets are strategic substitutes, and within each mirror pair the only symmetric equilibrium is a mixed lottery over who is exposed. Symmetric play survives at a single knife-edge budget where utility saturation flips the trough into a peak—the one symmetric pure equilibrium, and the welfare-best one. At every other pure equilibrium the players' realised risks are strictly unequal, heterogeneous preferences sort the exposed role onto whoever values privacy least, and the coalition attack of [5] is the coordinated limit of the same incentive. The phenomenon is a property of the sampling-based realisation—its sensitivity-based sibling, replayed through the identical game, has a unique symmetric equilibrium—and its practical bite tracks attack power: at the measurement floor the strategic pressure vanishes; against the strongest adversary the deployed mechanism admits—the level any risk assessment must price—it does not.

Limitations. Four boundaries of the analysis are worth stating plainly. First, the one-shot game assumes full information and simultaneous play; our repeated Bayesian extension (Section VI-C, Section F) covers a myopic follower and a learning leader, but the fully two-sided repeated game—a target who manages what their actions reveal [54]—and finite-horizon regret rates in the style of [53] remain open. Second, our players are silos choosing scalar budgets; per-record budget distributions within a silo, and the granularity question they raise, are untreated. Third, the continuous-action existence theorem covers the small-p regime that DP-SGD occupies, not the saturation corner; and on coarse grids the n-player symmetric game can pin fictitious play to the diagonal (Section V-A), so the $n \geq 3$ analogue of Proposition 9 is conjectural. Fourth, our single measured instance (Section V-D) sits at the LiRA measurement floor; a measured instance with above-floor attack power is the missing experiment between the analytical game and deployment reality.

Open questions. Beyond closing those gaps: does the Δ-restricted game of Proposition 17 admit a unique (ideally symmetric) equilibrium, and at what model-utility cost? Is there a budget-coordination mechanism that keeps sampling-based iDP's utility advantage while provably removing the externality—or is the sensitivity-based realisation's symmetric equilibrium the price of strategic neutrality? And what should a contributor-facing contract disclose: our aware-target result (Section VI-B) says the coupling itself is the single most valuable fact a contributor can know, which turns disclosure design into a security control.

Conclusion. Individual differential privacy promises that each contributor controls their own risk. Under the sampling-based realisation the promise fails not only against adversaries but at equilibrium: each contributor's realised risk is set by everyone else's rational play, invisibly to every individual contract. In the hierarchy (6), the contract is written at the mechanism-agnostic first level; the game plays out at the mechanism-specific second level—the true bound on what any adversary can extract from the deployed mechanism, which every player's choice moves and no contract states; and any deployed attack realises some fraction of that bound at the third. The remedy is not to police the players but to complete the contract—bound the externality itself, so that what each contributor is promised is what, whatever everyone else does, they get.

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APPENDIX A
MECHANISM-SOLVER CONSTRUCTION

The mechanism solution $(\sigma, p_1, \ldots, p_n)$ for a given action profile $\boldsymbol{\varepsilon}$ is obtained by a nested bisection. For each candidate σ, we find the largest per-group p_g that satisfies the I-fold (p_g, σ)-subsampled Gaussian at (ε_g, δ)-DP under (3). Because the achieved ε is monotone-increasing in p_g, a unique $p_g \in (0, 1]$ exists (capped at 1 if the budget cannot be exhausted); we locate it by binary search using the RDP accountant [14] until the achieved ε matches ε_g to a relative tolerance of

10^{-3}. We then bisect σ on top of this inner search, using the realised batch size $\hat{B}(\sigma) = \sum_g |G_g|\, p_g(\sigma, \boldsymbol{\varepsilon})$ as the outer-loop monitor, until $\hat{B}$ matches B to the same tolerance. Profiles in which any p_g falls below 10^{-5} are flagged infeasible per Assumption 4; below that threshold the group's contribution to (4) is numerically negligible and the iDP contract is trivially satisfied. The qualitative results are insensitive to this threshold across $[10^{-6}, 10^{-3}]$ (Section H).

Once (σ, p_g) is fixed, the realised MIA advantage of group g is computed exactly from the privacy-loss distribution of the I-fold (p_g, σ)-subsampled Gaussian [22] by evaluating the trade-off function and (5). This matches the formulation of [5] and yields the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B
PROOFS FOR SECTION IV

A. Proof of Lemma 7

We track how the solver absorbs one tightened budget. Everything rests on two hypothesis-testing facts about a single (p, σ) mechanism; the rest is bookkeeping through the two constraints. *(i) More sampling leaks strictly more:* for $p < p'$, the member hypothesis at rate p is a convex blend of the one at rate p' with the non-member hypothesis (with weight $1 - p/p'$ on the null), and blending an alternative toward the null makes every test strictly weaker. *(ii) More noise leaks strictly less:* at fixed p, the two hypotheses sit at mean shift $1/\sigma$, which shrinks as σ grows. Both orderings survive I-fold composition [21]; together they are amplification by subsampling [16]. Hence the budget a mechanism achieves at the shared δ, $\varepsilon(p, \sigma)$, is smooth, strictly increasing in p, and strictly decreasing in σ.

Now solve the constraints in order. Constraint (3) inverts fact (i): at a given σ, group g's binding rate $p_g(\sigma, \varepsilon_g)$ is the unique rate whose leakage exhausts the budget, and it rises in ε_g (a looser budget permits more sampling) and in σ (more noise creates sampling room, by (ii)). Constraint (4) then pins σ: the expected batch $F(\sigma) := \sum_g |G_g|\, p_g(\sigma, \varepsilon_g)$ is strictly increasing in σ, so exactly one σ delivers B.

Finally, tighten ε_j. At the old σ only p_j moves, downward, so the batch undershoots B; lowering σ would lower every rate and undershoot further, so the unique restoring direction is $\sigma \uparrow$. Implicit differentiation of $F(\sigma) = B$ makes this quantitative,

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;=\; -\,\frac{|G_j|\, \partial p_j / \partial \varepsilon_j}{F'(\sigma)} \;<\; 0,$$

and every other group's budget is unchanged, so its rate moves only through the noise: $\partial p_i / \partial \varepsilon_j = (\partial p_i / \partial \sigma)(\partial \sigma / \partial \varepsilon_j) < 0$. Note where the argument stops: if some p_g caps at 1 (a budget the noise cannot exhaust), (3) stops binding for that group and the signs can degenerate—the saturation corner excluded by the hypotheses and met once in Section V. □

B. Proof of Corollary 8

For a single step, the trade-off function of the subsampled pair is *exactly* the mixture $T_{p,\sigma}(\alpha) = (1-p)(1-\alpha) + p\, T_\sigma(\alpha)$: the likelihood ratio of the mixture against the base Gaussian is monotone in the same statistic as the unsubsampled pair, so the same tests are optimal ([16], Poisson-subsampling amplification). Its sup-gap—the advantage, which equals the total-variation distance of the pair—is therefore exactly $p\,\mathrm{Adv}_G(\sigma)$ per step, and total variation is additive to leading order under I-fold products:

$$\begin{aligned} A(p, \sigma) \;&=\; I\, p\, \mathrm{Adv}_G(\sigma) + O\big((Ip)^2\big), \\ \frac{\partial^2 A}{\partial p\, \partial \sigma} \;&=\; I\, \mathrm{Adv}'_G(\sigma) + O(I^2 p), \end{aligned}$$

the latter negative, valid in the regime $Ip \ll 1$ (the composed remainder is controlled by the product Ip, not by p alone). For the externality sign, substitute the small-p closed form $p_i = \sigma\sqrt{\rho_i}/\sqrt{2I\alpha}$ behind (12) into the chain rule for $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j$ and factor out $(\partial \sigma / \partial \varepsilon_j) < 0$, which yields (13). The bracket is the σ-derivative of $\sigma\,\mathrm{Adv}_G(\sigma)$ divided by σ; setting $u := 1/(2\sigma)$ and $f(u) := \Phi(u) - \Phi(-u) - 2u\phi(u)$ gives $f(0) = 0$ and $f'(u) = 2u^2\phi(u) > 0$, so the bracket is strictly positive and the leading $O(p_i)$ term of $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j$ is strictly negative. □

Intuitively, a tightened ε_j raises both p_i and σ together ($p_i \propto \sigma$ along the binding constraint): being sampled more exposes them, the extra noise protects them. The bracket of (13) is the net of those two channels along the line $p_i \propto \sigma$, and its positivity says the vulnerability added by the extra sampling strictly outweighs the protection added by the extra noise—at every σ, as long as p_i is small enough that their advantage is essentially linear in their sampling rate.

C. The full-region sign: reduction, obstruction, certificate

Proposition 18 (Full-region sign: reduction and certificate)**.** *Under the hypotheses of Lemma 7, with strict interior $p_g(\boldsymbol{\varepsilon}) \in (0, 1)$ for every group: (i) at fixed ε_i, the mechanism holds player i's leading-order (Rényi) privacy content invariant in ε_j, so the cross-effect on $A_{\mathrm{PLD},i}$ is carried entirely by higher-order terms of the privacy profile; (ii) the sign of that residual is $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j < 0$—analytically in the small-p regime (Corollary 8), and by numerical certificate on every feasible interior cell of the K=200 mechanism grid (Section D).*

Reduction: the leading-order effect is exactly zero. The composed Poisson-subsampled Gaussian trade-off function $T_{p,\sigma}$ is sandwiched between two Gaussian trade-off functions,

$$T_{\mathcal{G}(\mu_+(p,\sigma))}(\alpha) \;\le\; T_{p,\sigma}(\alpha) \;\le\; T_{\mathcal{G}(\mu_-(p,\sigma))}(\alpha), \quad \alpha \in [0, 1],$$

with scales $\mu_+(p, \sigma) = 2p\sqrt{I}/\sigma$ (Rényi-order-two → zCDP → GDP conversion; [14], [58]) and $\mu_-(p, \sigma) = p\sqrt{I}\sqrt{e^{1/\sigma^2} - 1}$ (Gaussian-DP central-limit envelope; [21], Cor. 3.3), and $\mathrm{Adv}(\mathcal{G}(\mu)) = 2\Phi(\mu/2) - 1$ strictly increasing in μ. Now observe what an ε_j-move does to player i's pair (p_i, σ) on the strict interior: their budget constraint (3) stays binding, so $p_i = \sigma\sqrt{\rho_i}/\sqrt{2I\alpha}$—the mechanism moves along the line $p_i \propto \sigma$, on which the ratio p_i/σ, and hence the upper-envelope scale μ_+, is *constant*. This is the structural content of part (i), and worth pausing on: the mechanism is built to hold every player's nominal guarantee fixed, so at the leading (Rényi-order-two) resolution the externality on $A_{\mathrm{PLD},i}$ is exactly invisible.

The entire phenomenon lives in the higher-order shape of the privacy profile—which is precisely why it hides *below* the contract, as slack, rather than ever violating it.

Obstruction: why no full analytic closure. Because the leading term cancels, the sign of $\partial A_{\mathrm{PLD},i}/\partial\varepsilon_j$ must come from the residual, and the envelope pair is too loose to extract it: along the line, μ_+ is constant while μ_- moves *against* the claimed direction (as σ falls, $\sigma\sqrt{e^{1/\sigma^2}-1}$ rises), so the sandwich brackets $A_{\mathrm{PLD},i}$ without ordering its response. Likewise the pointwise profile monotonicities—A increasing in p at fixed σ, decreasing in σ at fixed p [16]—pull in opposite directions along the curve and do not decide it either. In the small-p regime the residual *is* decided: Corollary 8 evaluates the derivative along the line exactly, and the bracket of (13) says the sampling channel strictly dominates the noise channel. Beyond small p we know of no analytic argument and flag the closure as open.

Certificate. The full-region claim is therefore certified exhaustively instead: the unit-stride finite-difference sign is negative on every feasible interior cell of the $K{=}200$ grid (Section D). The saturation corner ($p_g \to 1$, where (3) stops binding and p_i decouples from σ) and the feasibility edge (where the implicit differentiation of (12) is undefined) remain outside the hypotheses. □

D. Proof of Proposition 9

The proof exploits the mechanism's aggregator structure: the noise σ sees only the *sum* of the transformed budgets, so at a symmetric point everything the players share cancels between own- and cross-derivatives, and only the share term—each player's slice of the conserved batch—can decide the second-order test.

Setup. Change variables to $s_i := \sqrt{\rho(\varepsilon_i,\delta)}$ where $\rho(\cdot,\delta)$ is the RDP-order-α bound from the small-p closed form (12). Then $\varepsilon_i \mapsto s_i$ is a C^1 diffeomorphism on the interior; substituting the closed form of $p_g(\sigma,\varepsilon_g)$ into (4) yields, with $|G_1| = |G_2| = m$,

$$\sigma(s_1,s_2) \;=\; \frac{C}{s_1+s_2}, \qquad p_i(s_1,s_2) \;=\; \frac{C}{\sqrt{2I\alpha}}\cdot\frac{s_i}{s_1+s_2}, \tag{17}$$

with $C := B\sqrt{2I\alpha}/m > 0$. Utility becomes $u_i(s_1,s_2) = V(\sigma(S)) - \lambda\, A(p_i,\sigma(S))$ with $S := s_1+s_2$ and the same $A(\cdot,\cdot)$ for both players.

Diagonal FOC collapses. Fix $s_1 = s_2 = s > 0$, so $\sigma = C/(2s)$ and $p_1 = p_2 = C/(2\sqrt{2I\alpha}) =: p$. Write $a := \partial A/\partial p$ and $b := \partial A/\partial\sigma$ at (p,σ); by hypothesis $a > 0$. From (17), $\partial\sigma/\partial s_i = -\sigma/(2s)$, $\partial p_1/\partial s_1 = p/(2s)$, $\partial p_1/\partial s_2 = -p/(2s)$: player i's sampling rate is monotone in their *share* s_i/S. Player 1's first-order condition in s_1,

$$\frac{\partial u_1}{\partial s_1} \;=\; \big[V'(\sigma) - \lambda b\big]\Big(-\tfrac{\sigma}{2s}\Big) \;-\; \lambda a\,\tfrac{p}{2s} \;=\; 0, \tag{18}$$

coincides with player 2's by symmetric relabelling, so any diagonal candidate is characterised by the single scalar equation (18).

Cross-derivative is positive. Using (18) and the aggregator identities,

$$\frac{\partial u_1}{\partial s_2} \;=\; \frac{\partial u_1}{\partial s_1} + \lambda a\Big(\frac{\partial p_1}{\partial s_1} - \frac{\partial p_1}{\partial s_2}\Big) \;=\; \lambda a\cdot\frac{p}{s} \;>\; 0.$$

Second-order test. Apply the difference operator $D_1 - D_2$ to $D_1 u_1$. Because σ depends on (s_1,s_2) only through S, every term that is a function of S alone is killed: $V'(\sigma)\sigma'(S)$ drops out entirely. Two families survive, both through the p_1-dependence. First, in the term $-\lambda A_\sigma(p_1,\sigma)\,\sigma'(S)$ the coefficient A_σ depends on p_1, contributing $-\lambda\, a_{p\sigma}\,(p/s)\,\sigma'(S) = +\frac{\lambda p}{2s^2}\,\sigma\, a_{p\sigma}$ (using $\sigma'(S) = -\sigma/2s$). Second, the own-rate term $-\lambda A_p\, p_{1,1}$ contributes $-\lambda\, a_{pp}(p/s)\,p_{1,1} - \lambda\, a\,(D_1 - D_2)p_{1,1} = -\frac{\lambda p}{2s^2}\,p\,a_{pp} + \frac{\lambda p}{2s^2}\,a$ (from (17), $p_{1,1} = p/2s$ and $(D_1-D_2)p_{1,1} = -p/2s^2$). Altogether, at the diagonal,

$$\frac{\partial^2 u_1}{\partial s_1^2} - \frac{\partial^2 u_1}{\partial s_1\partial s_2} \;=\; \frac{\lambda\, p}{2\,s^2}\,\big(a + \sigma\, a_{p\sigma} - p\, a_{pp}\big), \tag{19}$$

with $a_{p\sigma} := \partial^2 A/\partial p\,\partial\sigma$ and $a_{pp} := \partial^2 A/\partial p^2$ at (p,σ). The difference alone does not decide the test: the trough requires $\partial^2 u_1/\partial s_1^2 > 0$ itself. Differentiating once more and evaluating at the diagonal—the $a_{p\sigma}$ contributions to the mixed partial cancel—gives

$$\frac{\partial^2 u_1}{\partial s_1\,\partial s_2} \;=\; \frac{1}{4s^2}\Big[\,\sigma^2 V'' + 2\sigma V' + \lambda\big(p^2 a_{pp} - \sigma^2 A_{\sigma\sigma} - 2\sigma A_\sigma\big)\Big], \tag{20}$$

and adding (19) yields the exact diagonal curvature,

$$4s^2\,\frac{\partial^2 u_1}{\partial s_1^2} \;=\; \underbrace{\sigma^2 V'' + 2\sigma V'}_{\text{shared utility}} \;+\; \lambda\,\underbrace{\big(2pa + 2p\sigma a_{p\sigma} - p^2 a_{pp} - \sigma^2 A_{\sigma\sigma} - 2\sigma}_{\text{own risk}} \tag{21}$$

A diagonal first-order candidate is a strict local *minimum* of own utility—a trough, hence no equilibrium—iff $\Psi > 0$, and a strict local *maximum*, a symmetric equilibrium candidate, iff $\Psi < 0$.

The provable regime. In the small-p expansion, $a = I\mathrm{Adv}_G$, $a_{p\sigma} = I\mathrm{Adv}'_G$, $A_\sigma = Ip\mathrm{Adv}'_G$, $A_{\sigma\sigma} = Ip\mathrm{Adv}''_G$, $a_{pp} = O(I^2 p)$, so the risk term collapses to $\lambda Ip\,[\,2\mathrm{Adv}_G(\sigma) - \sigma^2\mathrm{Adv}''_G(\sigma)\,] + O((Ip)^2)$, which is strictly positive: with $u := 1/(2\sigma)$, $\sigma^2\mathrm{Adv}''_G = 4u\varphi(u) - 2u^3\varphi(u)$, so $2\mathrm{Adv}_G - \sigma^2\mathrm{Adv}''_G = 2\,[\,2\Phi(u) - 1 - 2u\varphi(u)\,] + 2u^3\varphi(u)$, and the bracket vanishes at $u = 0$ with derivative $2u^2\varphi(u) > 0$. For the canonical $V(\sigma) = 1/(1+\sigma^2)$ the shared-utility term is

$$\sigma^2 V'' + 2\sigma V' \;=\; \frac{2\,\sigma^2(\sigma^2-3)}{(1+\sigma^2)^3},$$

non-negative exactly on $\sigma \ge \sqrt{3}$—the complement of the strategic-substitutes region $\sigma \le \sqrt{3}$ of Theorem 11. On $\sigma \ge \sqrt{3}$ both terms of Ψ are positive, the candidate is a trough, non-FOC diagonal profiles fail the first-order test directly, and the C^1 diffeomorphism transports the conclusion back to ε-coordinates. □

Below $\sigma = \sqrt{3}$ the two terms of Ψ compete: the shared utility saturates—V is bounded, so joint loosening buys less and less—and its curvature along the diagonal turns negative, while the risk term still pushes toward the trough. Where the

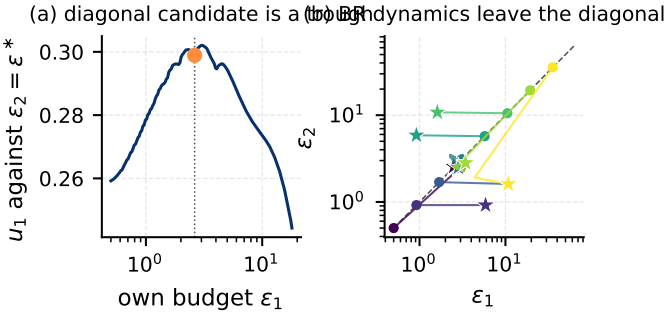


Fig. 13: **Anatomy of symmetry breaking** ($K = 200$ mechanism cache, symmetric two-player game). *(a)* Player 1's utility against an opponent fixed at the diagonal first-order candidate ε^*: the symmetric profile (dot) is a local minimum of their own utility—deviating in either direction pays. *(b)* Alternating best-response dynamics started at diagonal profiles (circles): every trajectory leaves the diagonal at the first response and terminates at an off-diagonal pure Nash equilibrium (stars).

utility saturation wins, the diagonal candidate flips from trough to peak and a symmetric equilibrium becomes possible. Two caveats bound what the model analysis can claim there. First, the sign of Ψ below $\sqrt{3}$ is parameter-dependent; it must be measured. Second, the aggregator form (17) is itself the small-p closed form, and the canonical operating points sit at $Ip \approx 5$, where the composed advantage saturates and the expansion is quantitatively unreliable—the difference bracket in (19), for instance, evaluates *negative* at the operating points (-2.1 at $\varepsilon = 2.96$, -4.9 at $\varepsilon = 10.28$, by finite differences of the exact PLD advantage), so no positivity claim for it survives outside the expansion. We therefore classify the exact mechanism's diagonal directly, on the clean $K = 200$ cache: every interior feasible diagonal cell admits a strictly profitable unilateral deviation *except* in a single window around $\varepsilon \approx 3$, where the first-order candidate at $\varepsilon^\star = 2.96$ ($\sigma = 1.16 < \sqrt{3}$) is a strict local maximum of own utility—confirmed on a 25-point local refinement of the action grid, on which every deviation strictly loses—and a global best response to itself: the symmetric pure equilibrium reported in Section IV-B (Section B-E).

E. Numerical anatomy of symmetry breaking

The quantities behind the discussion around Proposition 9, all computed on the canonical $K = 200$ mechanism cache at $\lambda = 1$ and summarised in Fig. 13.

Diagonal trough and the saturation window. Outside the saturation window, every feasible diagonal cell admits a strictly profitable unilateral deviation, with median best-deviation gain 0.122 utility across the 200 diagonal cells (Fig. 13(a)). The discrete own-action first-order condition changes sign at eight interior diagonal cells, $\varepsilon^* \in [1.78, 4.07]$; at the lower candidates the trough holds (e.g. $\varepsilon^* = 2.11$: $+0.007$ upward, $+0.0003$ downward), and at $\varepsilon^* = 2.96$ it flips—both deviations lose, the cell is the symmetric pure equilibrium of Section IV-B.

Strategic substitutes. The discrete cross-difference $u_1(\varepsilon_1^{+h}, \varepsilon_2^{+h}) - u_1(\varepsilon_1^{+h}, \varepsilon_2^{-h}) - u_1(\varepsilon_1^{-h}, \varepsilon_2^{+h}) + u_1(\varepsilon_1^{-h}, \varepsilon_2^{-h})$ centred on diagonal cells is negative on 81.2% / 90.2% / 95.2% of cells at strides $h = 4$ / 8 / 16; the coarser strides average over the PLD-accountant's discreteness and carry the signal.

Best-response dynamics. Alternating best-response dynamics started at each of the 200 feasible diagonal profiles end off-diagonal in 197 cases—the three exceptions start inside the saturation window and stay at the symmetric equilibrium—with a median of two steps of play; 134 of the 200 runs terminate at a pure Nash equilibrium within the iteration budget, the remainder in best-response cycles between near-tied asymmetric profiles (Fig. 13(b)).

Symmetric equilibria of the restriction. Restricted to the two budgets of the mirror pair $(1.12, 7.13)$, the game is Hawk–Dove: $u(\text{tight}, \text{loose}) = (0.505, 0.253)$, both symmetric profiles of the restriction unstable, and the restriction's unique symmetric equilibrium mixed, playing tight with probability 0.445 at value 0.349—strictly between the two role payoffs. On the full $K = 22$ game, exact support enumeration finds no proper mixed symmetric equilibrium up to support size five; the game's symmetric equilibrium is the pure one of the saturation window, present already at that resolution.

F. Proof of Proposition 10

Write $f(x, y)$ for player i's realised advantage at own budget x and opponent budget y; by $|G_1| = |G_2|$ the noise satisfies $\sigma(x, y) = \sigma(y, x)$ and f is common to both players. The exchange condition referenced in the main text is

(S) for every $x > y$ in the feasible region, $\big[V(\sigma_H) - V(\sigma_M)\big]\big[f(x, y) - f(y, y)\big] \geq \big[V(\sigma_M) - V(\sigma_L)\big]\big[f(x, x) - f(y, x)\big]$,

where $\sigma_L = \sigma(y, y) > \sigma_M = \sigma(x, y) > \sigma_H = \sigma(x, x)$ by Lemma 7.

Fix a pure NE interior to the feasible region and suppose for contradiction $\varepsilon_1^\star > \varepsilon_2^\star$; write $x = \varepsilon_1^\star$, $y = \varepsilon_2^\star$. With $V' < 0$, both $\Delta V_H := V(\sigma_H) - V(\sigma_M)$ and $\Delta V_L := V(\sigma_M) - V(\sigma_L)$ are strictly positive, and by own-budget monotonicity the lifts $D_1 := f(x, y) - f(y, y)$ and $D_2 := f(x, x) - f(y, x)$ are strictly positive. Player 1 not deviating from x to y gives $\lambda_1 D_1 \leq \Delta V_L$; player 2 not deviating from y to x gives $\lambda_2 D_2 \geq \Delta V_H$. Dividing,

$$\frac{\lambda_2}{\lambda_1} \geq \frac{\Delta V_H}{\Delta V_L} \cdot \frac{D_1}{D_2} \geq 1$$

by (S), contradicting $\lambda_1 > \lambda_2$. □

Sufficient sub-conditions for (S): V midpoint-convex on $[\sigma_H, \sigma_L]$ together with f submodular on the $\{y, x\}^2$ block (which closes in the high-noise regime $\sigma > \sqrt{3}$ via (12)); or pointwise numerical verification at the equilibrium candidate, which is how the $K = 200$ sweeps of Section V are certified. At $\lambda_1 = \lambda_2$, (S) holds with equality and the argument degenerates to closure of the NE set under the swap $(\varepsilon_1, \varepsilon_2) \leftrightarrow (\varepsilon_2, \varepsilon_1)$—the mirror pairs of Section IV-B.

G. Proof sketch of Theorem 11

Expand $\partial^2 u_i/(\partial\varepsilon_i\partial\varepsilon_j)$ by the chain rule. For the shared-V contribution, use the aggregate form (12): writing S for the weighted budget aggregate, $\sigma_i\sigma_j = \sigma^2 S_i S_j/S^2$ and $\sigma_{ij} = 2\sigma S_i S_j/S^2$, so

$$V''(\sigma)\,\sigma_i\sigma_j + V'(\sigma)\,\sigma_{ij} \;=\; \sigma_i\sigma_j\Big[V''(\sigma) + \tfrac{2V'(\sigma)}{\sigma}\Big],$$

which is ≤ 0 exactly when $V''+2V'/\sigma \le 0$—for the canonical $V = 1/(1+\sigma^2)$, when $\sigma \le \sqrt{3}$. The $-\lambda_i A_{\mathrm{PLD},i}$ contribution's dominant terms in the small-p regime are $-\lambda_i A_{i,p\sigma}\,(p_{i,i}\sigma_j + p_{i,j}\sigma_i)$ with $A_{i,p\sigma} = I\mathrm{Adv}'_G(\sigma) + O(I^2 p_i) < 0$ (Corollary 8), and carry the sign of the strictly negative bracket of (13). Hence $\partial^2 u_i/(\partial\varepsilon_i\partial\varepsilon_j) \le 0$ on the box $\mathcal{B}$: the budgets are strategic substitutes. For two players, reversing one player's coordinate turns decreasing differences into increasing differences, and a box is a compact lattice under any coordinatewise order, so Milgrom–Roberts ([49], Thm. 5) delivers a non-empty complete lattice of pure NEs of the box-restricted game with extremal elements; equilibria of the restriction whose unrestricted best responses stay inside $\mathcal{B}$ are equilibria of the PEG. The one-coordinate flip is unavailable for $n > 2$ (substitutes cannot become complements for all pairs at once), which is why the theorem is stated for two players. Roughly a quarter of interior cells of the *full* feasible region fail the numerically tested substitutes sign, so the box qualifier is not removable with this proof strategy. □

H. Proof sketch of Proposition 12

The obstruction is that the externality enters each utility weighted by the player's own λ_i and the *partner's* group size; a potential would require these weights to reconcile symmetrically across every pair, which the coupling forbids.

Cardinal potentials. Implicit differentiation of (12) gives $\sigma_{ij} = 2\sigma\,|G_i||G_j|(r')^2/(rN)^2$ at a symmetric-budget reference, where $r := \sqrt{\rho}$ and $N := \sum_g |G_g|$; the shared-V contribution to the Monderer–Shapley cross-partial condition $\eta_j\,\partial^2 u_i/\partial\varepsilon_i\partial\varepsilon_j = \eta_i\,\partial^2 u_j/\partial\varepsilon_i\partial\varepsilon_j$ is symmetric under $i \leftrightarrow j$ and cancels. Expanding the $A_{\mathrm{PLD},i}$ contribution reduces the residual to $\eta_j\lambda_i|G_j|(2|G_i| - N) = \eta_i\lambda_j|G_i|(2|G_j| - N)$ at every symmetric profile, which has no positive solution (η_i, η_j) for $|G_i| \ne |G_j|$; in the fully symmetric case the residual carries $(\eta_j - \eta_i)\,d^2V$, which is nonconstant along the feasible interval and cannot vanish uniformly unless $\eta_1 = \eta_2$, in which case the heterogeneous case above bites.

Ordinal potentials. By [31], Lemma 2.5, an ordinal potential precludes strict improvement cycles. On the $K = 22$ discretisation of the canonical game, the cycle $(1.587, 5.040) \to (2.000, 5.040) \to (2.000, 4.000) \to (1.587, 4.000) \to (1.587, 5.040)$ has each step strictly improving for the deviating player (unilateral gains 7.3×10^{-7}, 1.5×10^{-3}, 3.0×10^{-3}, 2.8×10^{-3}, re-verified from the mechanism cache). One caveat for the record: the smallest of the four gains sits near the accountant's numerical resolution; the cycle's existence does not hinge on it alone—three of the four steps exceed 10^{-3}—but a fully tolerance-robust cycle would re-solve the four profiles at finer PLD discretisation. □

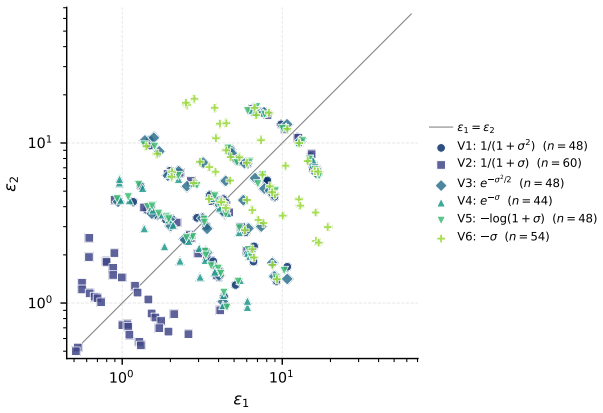


Fig. 14: **Model-utility ablation** ($K = 200$, $\lambda = 1$). Pure NE locations under six admissible shapes for V. Under every shape, 100% of the pure NEs sit off the diagonal.

Appendix C
Validity and role of V

[43] prove that for a 1-Lipschitz strongly-convex loss, the excess empirical risk of DP-ERM satisfies $\mathbb{E}[\hat{L}(\widehat{\theta}) - \hat{L}(\theta^*)] = O\big(d\log(1/\delta)/(n^2\varepsilon^2)\big)$; under the moments accountant of [7] this translates into a $\sigma^2 d/n^2$ dependence. Treating model utility as the negative of excess risk and renormalising to $[0, 1]$ gives $V(\sigma) \propto 1/(1+\alpha\sigma^2)$ for a problem constant α; we set $\alpha = 1$. The bound assumes convexity; the empirical DP-SGD literature [3], [7], [44] confirms the same qualitative direction in deep learning. The following pair, referenced from Section IV, confines V's analytical role to $\operatorname{sgn} V'$.

Assumption 19 (Admissible model-utility function)**.** $V\colon \sigma(\mathcal{F}) \to \mathbb{R}$ is *admissible* iff it is continuous, strictly decreasing, and bounded on the feasible noise range.

Proposition 20 (Qualitative invariance in V)**.** *For every admissible V: (i) the externality signs of Lemma 7, Corollary 8, and Proposition 18 hold; (ii) Corollary 13 holds; and (iii) at any interior best response,* (16) *depends on V only through* $\operatorname{sgn} V'$.

The proof is by inspection: the mechanism constraints (3)–(4) and the closed form $p_g(\sigma, \varepsilon_g)$ never read V, so Lemma 7, Corollary 8, and Proposition 18 and Corollary 13 (which uses only the externality sign on the target's coordinate) are V-free; differentiating (7) in ε_i gives (16), whose sign structure reads V only through $\operatorname{sgn} V'$; and continuity plus boundedness make the discretised game well-defined for every admissible V.

A. Numerical ablation over admissible shapes

We re-enumerate the canonical $K = 200$ equilibria under six admissible shapes: the canonical $V_1(\sigma) = 1/(1+\sigma^2)$, the slow-decaying $V_2 = 1/(1+\sigma)$, the Gaussian $V_3 = e^{-\sigma^2/2}$, the exponential $V_4 = e^{-\sigma}$, the concave $V_5 = -\log(1+\sigma)$, and the affine $V_6 = -\sigma$. Because the mechanism grid is V-free, each shape costs only $O(K^2)$ elementwise work on the cached tensor.

tag	shape	#NE	$\mathrm{mean}(b \to a)$	$\mathrm{agree}_{\mathrm{sign}}$
V_1	$1/(1+\sigma^2)$	48	0.000	100.0%
V_2	$1/(1+\sigma)$	60	0.075	100.0%
V_3	$e^{-\sigma^2/2}$	48	0.013	100.0%
V_4	$e^{-\sigma}$	44	0.062	100.0%
V_5	$-\log(1+\sigma)$	48	0.000	100.0%
V_6	$-\sigma$	54	0.111	100.0%

TABLE II: Per-V pure-NE count, mean one-sided Hausdorff distance from a V_1-NE to its nearest NE under V_k (decades of $\log_{10}\varepsilon$), and externality-sign agreement with V_1 over interior feasible edges. $K = 200$, $\lambda = 1$.

Across the six shapes the pure-NE count stays in 44–60 and the NE set remains entirely off-diagonal (Fig. 14, Table II); each V_1-equilibrium has a match within 0.11 decades under every alternative; and the externality-sign pattern over feasible edges agrees with V_1 on 100% of edges— the numerical fingerprint of the additive separation of (7), in which the sign of $\partial u_i/\partial \varepsilon_j$ is mechanism-determined and V-free.

Appendix D
The externality response and its sign certificate

The externality in isolation. The equilibrium results all flow through one primitive: the response of a target's realised advantage to the others' budgets. We isolate it by fixing the target's budget $\varepsilon_T \in \{4, 16\}$ and sweeping the other group's budget ε_O and dataset share (Fig. 15). The dependence is smooth and monotone, exactly as Proposition 18 predicts: at $\varepsilon_T = 4$ the target's advantage runs from 0.20 (others at $\varepsilon_O = 0.5$) to near zero (others at 64); at $\varepsilon_T = 16$, from 0.55 down to 0.05. The curves for different shares cross at $\varepsilon_O = \varepsilon_T$, where the others' contribution to σ matches the target's own; below the crossover a larger foreign share amplifies the externality, above it, suppresses it.

Finite-difference certificate. This is the certificate that part (ii) of Proposition 18 relies on beyond the small-p regime (Section B-C): we certify the sign on the $K = 200$ mechanism cache by finite differences. Of the 40,000 grid cells, 64.5% are feasible, matching the analytic participation frontier cell-for-cell. At unit stride—one grid step, no smoothing—the predicted cross-sign $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j < 0$ holds at every one of the 25,612 feasible interior differences, and the own-direction sign $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_i > 0$ likewise at 25,612 of 25,612. The certificate is exhaustive on the feasible region; no stride averaging or jitter exclusion is required.

Appendix E
Symmetric weight sweep

Sweeping the *common* weight $\lambda \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ for both players moves how much everyone cares, without breaking symmetry (Fig. 16). On the utility-leaning half of the sweep the asymmetric structure is insensitive: $\lambda \in \{0.125, 0.25, 0.5\}$ yields 34–62 pure NEs, all off-diagonal, and $\lambda = 1$ adds the saturation-window equilibrium to its 42 asymmetric ones. The privacy-heavy extreme is different in kind: at $\lambda = 2$ the grid has *no* pure equilibrium—only mixed play survives—and at $\lambda = 4$ a single corner equilibrium $(0.5, 0.5)$ remains. The weight also moves *where* the cloud sits (privacy-heavy play compresses budgets downward) and how costly equilibrium play is in aggregate, quantified in Section V-C: at $\lambda \le 0.25$ *every* equilibrium inflates aggregate advantage over the uniform baseline (up to +165%), at $\lambda = 0.5$ still 94% of them do, while at $\lambda = 1$ only 14% do and the $\lambda = 4$ corner deflates it by 82%.

Appendix F
The Bayesian Repeated PEG in full

Position in the literature. The classical framework for repeated play under one-sided private information is [54], whose central question is how the informed player manages what they reveal. The security-motivated strand has a leader committing each round against a follower of unknown type, refining a belief over the follower's payoff from observed responses [51]–[53]; [55] supply the posterior-consistency backbone. Selling privacy at heterogeneous private valuations [35] and Bayesian persuasion [59] are conceptually adjacent but solve different design problems. The BR-PEG of Section VI-C is the Stackelberg-learning template with the tabular follower payoff replaced by the DP-accountant coupling of Definition 1; the follower's private type is $\theta_H = (\lambda_H, n_H)$, the leader observes (ε_H^t, u_H^t) after each round, and the finite type space with a full-support prior makes Kalai–Lehrer merging automatic—the interesting content is what the leader does once the posterior concentrates.

Full proof of Proposition 15. Write $\bar{G}(\varepsilon_M; \pi) := \mathbb{E}_{\theta\sim\pi} G(\varepsilon_M; \theta)$ for the posterior-averaged score, a finite mixture of per-type scores, continuous in π in total variation and defined on the finite grid $\mathcal{A}$.

Step 1 (posterior collapse). By identifiability (I) and [55], $\pi^t \to \delta_{\theta_H^\star}$ π^0-a.s.: any type producing a distinct best response at the leader's action contributes strictly positive KL divergence per round, and divergences accumulate.

Step 2 (uniform score convergence). G is bounded in $[0, 1]$, so for every arm, $|\bar{G}(\varepsilon_M; \pi^t) - G(\varepsilon_M; \theta_H^\star)| \le 1 - \pi^t(\theta_H^\star) \to 0$, uniformly over the finite $\mathcal{A}$.

Step 3 (argmax lock-in). Let $\gamma > 0$ be the truth-argmax gap from (A) (not to be confused with the contract parameter Δ). Once $1 - \pi^t(\theta_H^\star) < \gamma/2$, for any $\varepsilon_M \neq \varepsilon_M^\star$,

$$\bar{G}(\varepsilon_M^\star; \pi^t) - \bar{G}(\varepsilon_M; \pi^t) \;\ge\; \gamma - 2\big(1 - \pi^t(\theta_H^\star)\big) \;>\; 0,$$

so the posterior-greedy arm equals $\varepsilon_M^\star$ for all subsequent rounds; separation from $\varepsilon^{\min}$ is (A). □

Clean demo instance. With three well-separated types ($\Theta = \{\lambda_H \in \{0.5, 1, 2\}\}$, truth $\lambda_H^\star = 2$, $n_H = n_M = 5000$, canonical parameters), the posterior collapses after a single round; from $t = 1$ the Bayesian-greedy leader plays $\varepsilon_M^\star \approx 3.08$, eliciting $\varepsilon_H \approx 19$ and $A_H \approx 0.454$ per round—roughly $2\times$ the naive constant attack $\varepsilon_M = 1$ ($A_H \approx 0.22$) and $\approx 50\times$ the

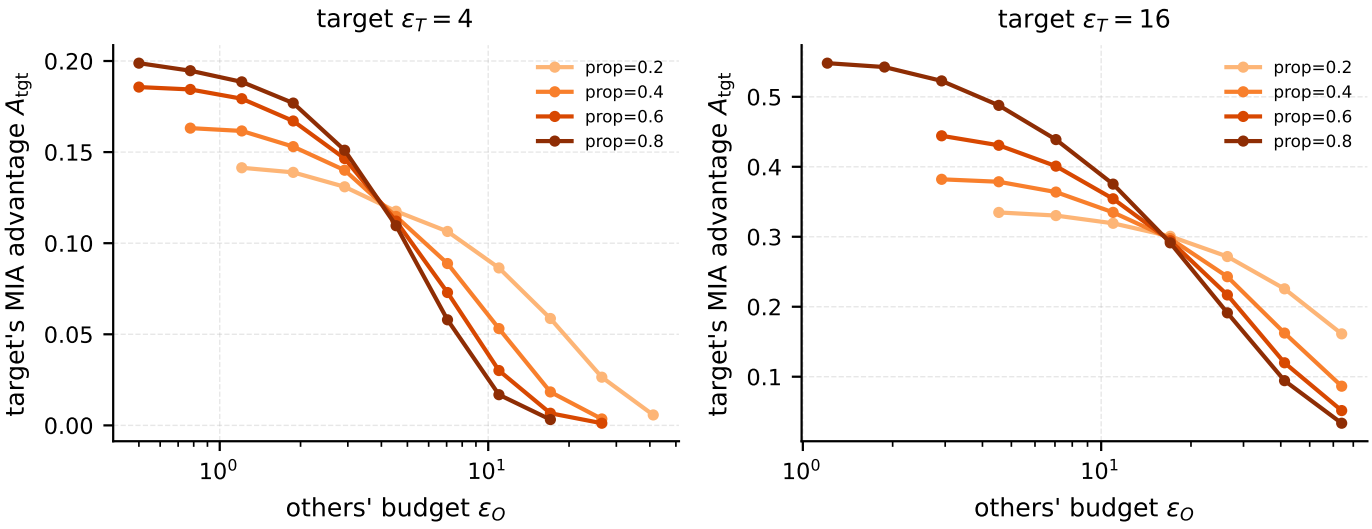


Fig. 15: **The externality in isolation.** Target's realised MIA advantage as a function of the other group's budget ε_O and dataset share, at fixed target budgets $\varepsilon_T \in \{4, 16\}$. The dependence is monotone in ε_O; curves cross at $\varepsilon_O = \varepsilon_T$, with larger foreign shares amplifying the externality below the crossover and suppressing it above.

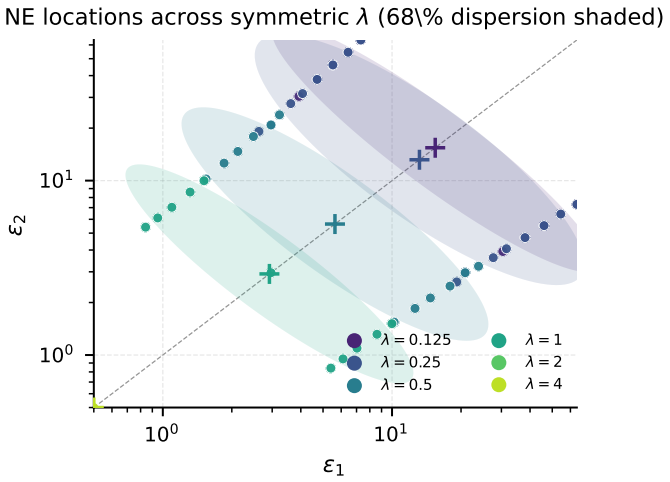


Fig. 16: **Symmetric weight sweep** ($K = 200$). Pure NE clouds at common weights $\lambda \in \{0.125, \ldots, 4\}$ (centroids and 68% dispersion ellipses); the dashed line is the diagonal. The NE set stays off-diagonal at every weight except the $\lambda = 1$ saturation-window equilibrium and the $\lambda = 4$ corner $(0.5, 0.5)$; at $\lambda = 2$ no pure equilibrium exists. The cloud shifts toward lower budgets as privacy weight grows.

benign baseline $\varepsilon_M = 64$ ($A_H \approx 0.01$) (Fig. 17). Assumption (A) is verified on the mechanism grid: the argmax is interior (≈ 2.1 above the naive anchor), because the leader's gain comes from eliciting a more exposing best response, not from forcing σ down. The regret instance of Fig. 12 hardens this demo—five overlapping types whose best responses coincide at several arms and observation noise $\sigma_{\text{obs}} = 0.15$ on u_H—so that identification takes ≈ 10 rounds rather than one and the regret separation between correct-prior, misspecified-prior, and prior-free attackers becomes visible.

What remains open. A strategising target—one who manages what their responses reveal, the two-sided game of [54]—and finite-horizon regret rates in the style of [52], [53] both fit the framework without modification and are left open.

Appendix G
Non-strategyproofness and the type-channel design problem

Environment. Player i's group size is public; their type is their privacy weight $\lambda_i > 0$, private. A direct-revelation iDP mechanism maps reported budgets to a mechanism solution satisfying (3)–(4), with no transfers; sampling-based iDP as deployed is the *identity mechanism* on the budget channel.

Proof of Proposition 16. Fix i and $\lambda_i > 0$. Choose opponent profiles on opposite sides of the externality: $\hat{\varepsilon}^{\text{L}}_{-i}$ loose, $\hat{\varepsilon}^{\text{T}}_{-i}$ tight. By Lemma 7 and the certified sign of Proposition 18, pointwise on Σ_i: $\sigma^{\text{T}} > \sigma^{\text{L}}$ and $A^{\text{T}}_{\text{PLD},i} \geq A^{\text{L}}_{\text{PLD},i}$, and the marginal utilities $\partial_{\hat{\varepsilon}_i} u_i^K = V'(\sigma^K)\partial_{\hat{\varepsilon}_i}\sigma^K - \lambda_i \partial_{\hat{\varepsilon}_i} A^K_{\text{PLD},i}$ differ across $K \in \{\text{L}, \text{T}\}$, so the best-response maps differ. A numerical witness in a scaled two-player instance ($|G_1| = |G_2| = 25{,}000$, $B = 128$, $I = 1953$, $\delta = 10^{-12}$, $\lambda_1 = 1$): against $\hat{\varepsilon}_2 = 4.0$ the best report is $\hat{\varepsilon}_1 \approx 2.92$; against $\hat{\varepsilon}_2 = 0.5$ it is ≈ 1.21. Sweeping $\hat{\varepsilon}_2$ over the grid, the best-response report takes six distinct values, generally rising with $\hat{\varepsilon}_2$. No single report dominates. □

Why the classical escapes fail. The large-market recovery of [57] requires single-agent influence to vanish; a silo moves (4) at $\Theta(1)$. The impossibility results for unbounded privacy valuations [35], [36] are not the obstacle (utility here is bounded); the obstacle is the coupled σ: the positive results of [37] model privacy costs that vanish when a report has no

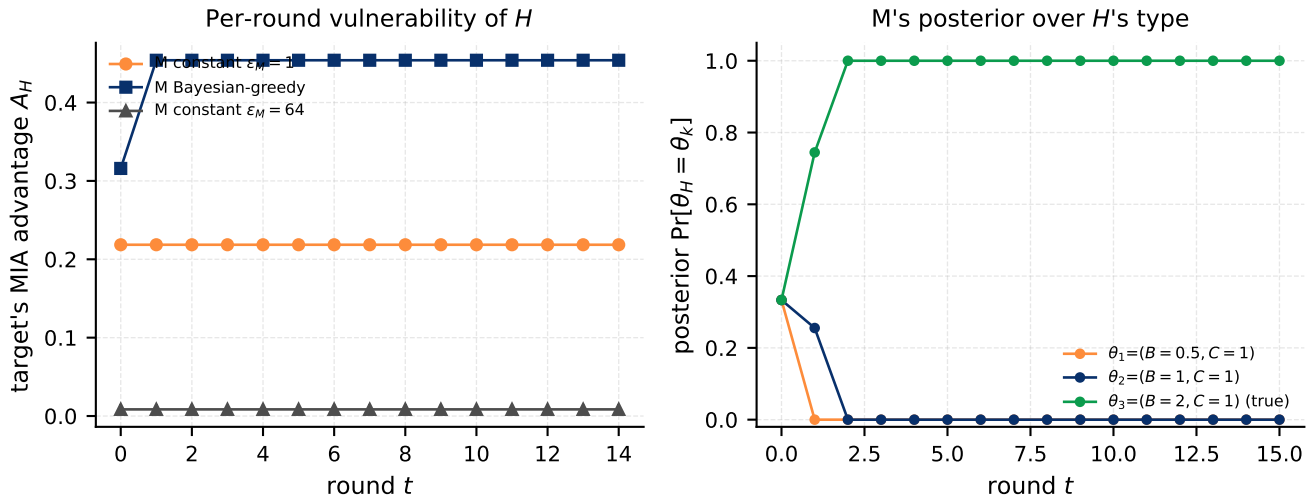


Fig. 17: **BR-PEG demo** (three well-separated types). Left: per-round realised advantage A_H under three leader policies. Right: the leader's posterior over θ_H under the Bayesian-greedy policy; it collapses to the truth within one round.

influence on the outcome, whereas in the PEG every report moves σ.

The type-channel problem. A repair would elicit λ_i rather than ε_i and allocate budgets centrally. With transfers, the envelope route reduces Bayesian incentive compatibility to monotonicity of the interim advantage $\overline{A}_i(\lambda_i)$ in the reported type [60]–[62]; three features keep the full theorem open here: the mechanism must choose one shared σ; the interim advantage couples all opponents' types through the joint solve; and there is no monetary transfer, leaving excess realised vulnerability as the only in-kind instrument—itself the quantity the contract constrains. We leave the constructive design problem to future work.

Appendix H
Sensitivity Checks

Grid refinement from $K = 22$ through $K = 200$ multiplies the resolved pure-NE count ($8 \rightarrow 48$) while preserving the structural invariants—zero diagonal equilibria, mirror pairing, the budget-ratio band, and the BR-loci geometry (Section IV-B). Substituting admissible alternatives for V preserves the structure (Section C-A). On a $K' = 50$ subgrid of the canonical game, the equilibrium count is insensitive to *lowering* the participation threshold (12 pure NEs at both 10^{-5} and 10^{-7}, over identical feasible regions) and drops to 6 only when the threshold is raised to 10^{-3}, which prunes boundary equilibria while leaving interior equilibria and all reported signs unchanged; the externality-sign certificate improves slightly under the stricter threshold (95.5% at 10^{-3} versus 92.7% at 10^{-5} at matched stride). Sweeps over δ, batch size and epoch count, and the accountant's Rényi orders are deferred to follow-up runs.